

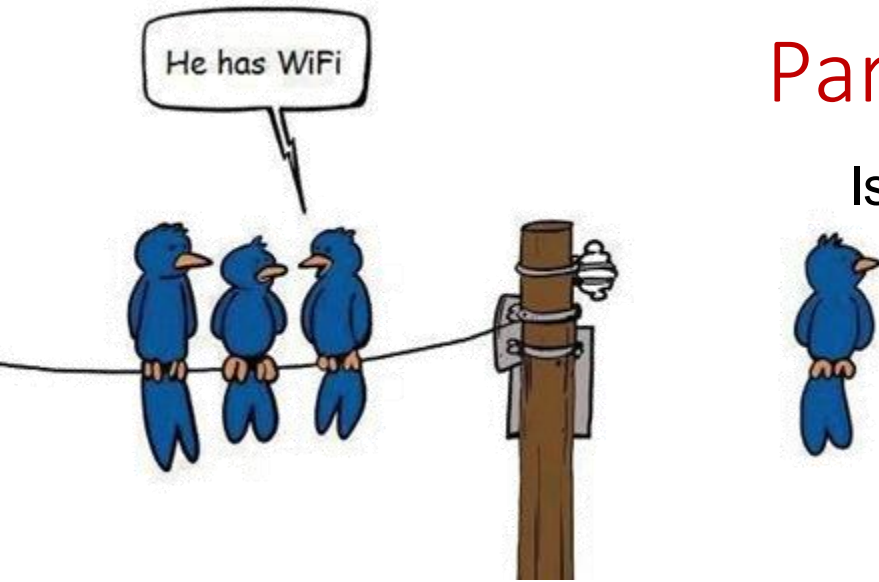
ECSE 4560/ 6560

Modern Communication Systems

Lecture 5: Wireless Systems Overview

Part 5 - Channel

Ish Jain, Fall 2025



Announcements

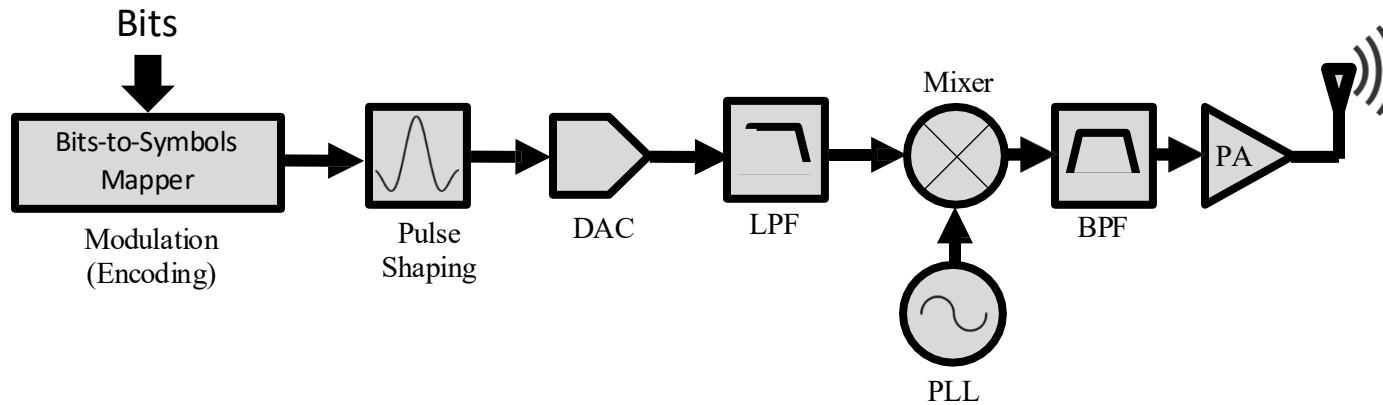
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Channel**
- Projects
 - A list of papers will be posted
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Proposal due 9/26
- HW1/Lab1 due Monday 9/22
 - Submit on Gradescope – Mark pages for each question parts
 - Matlab coding exercises – “Use Publish option to submit your code”
- HW2/Lab2 to be posted this weekend, due Mon 9/29

Today's Class: Wireless Overview 5

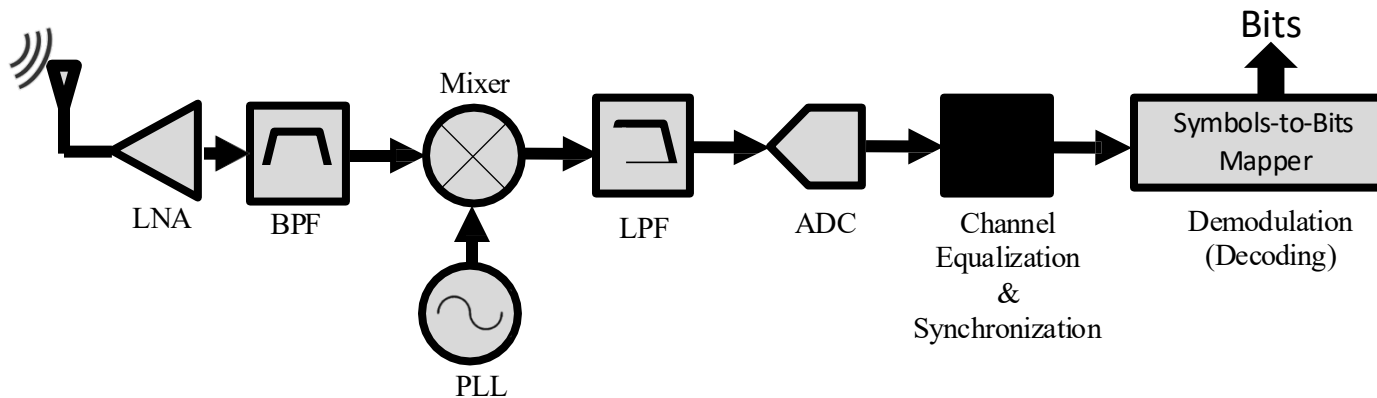
- QPSK BER (Review)
- Ideal Sampling and Reconstruction (Review)
- Practical ADC/DAC Sampling and reconstruction
- Wireless Channel

Wireless Communication System

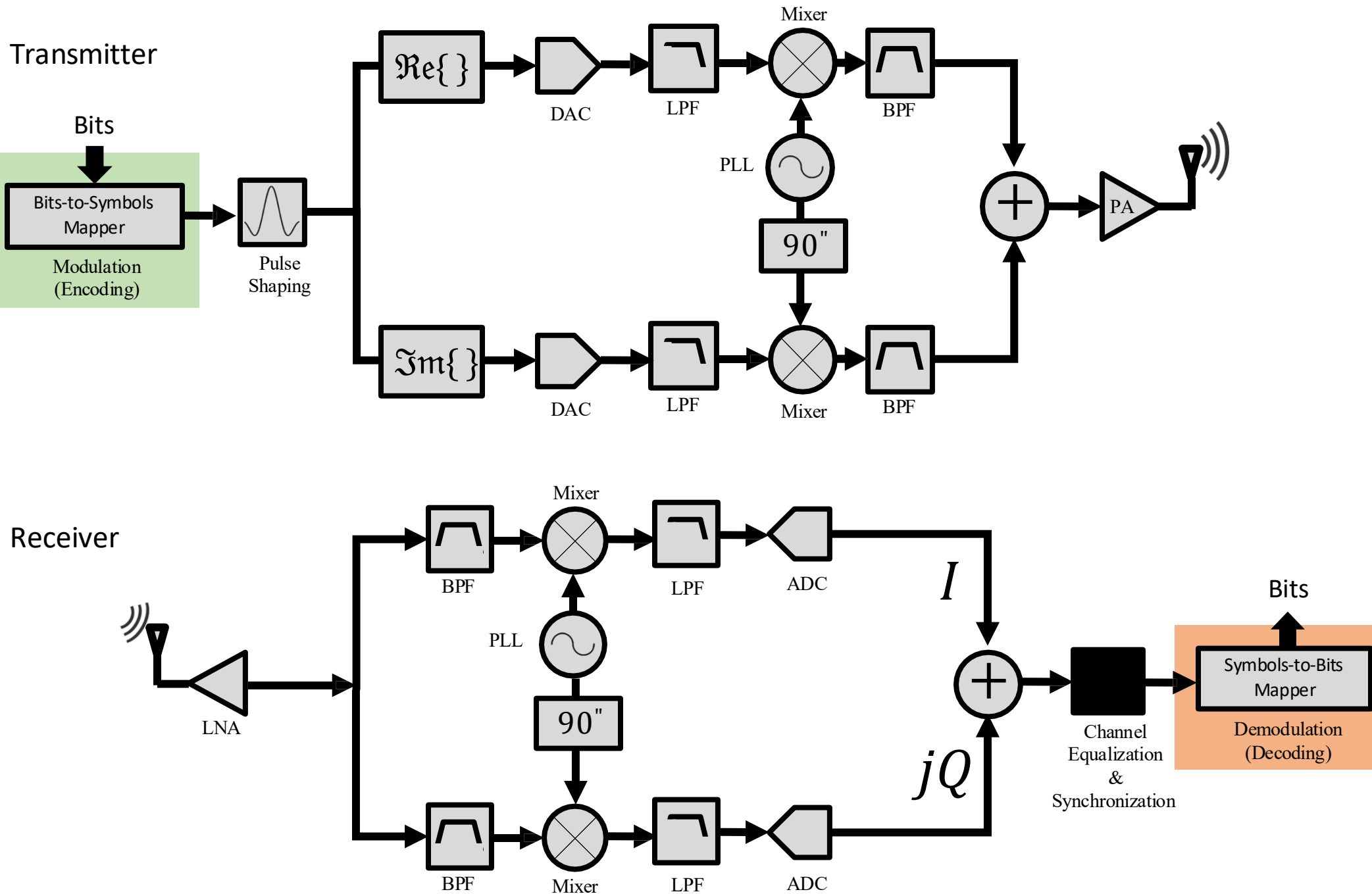
Transmitter



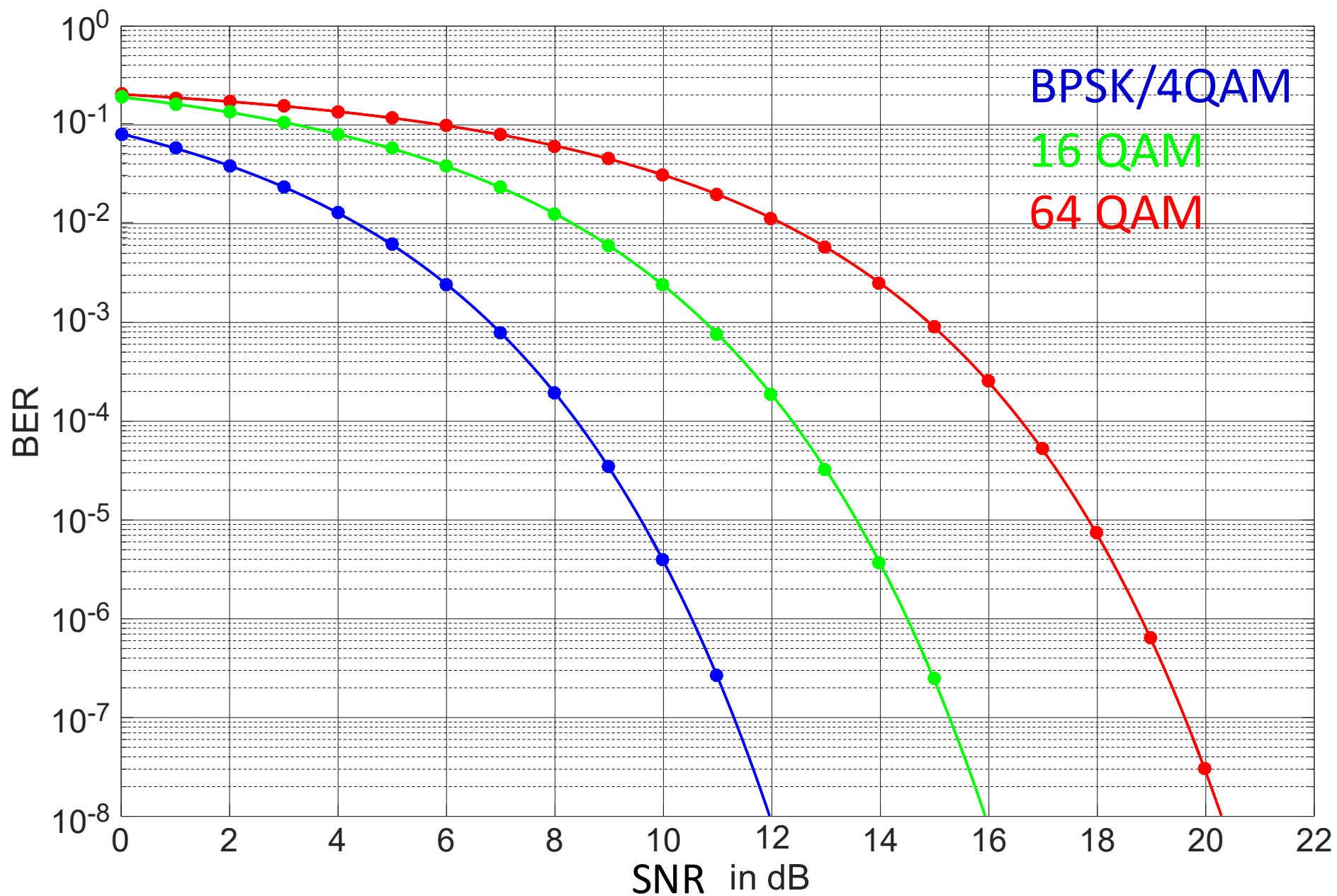
Receiver



Wireless Communication System



Bit Error Rate (BER)



Bit Error Rate (BER)

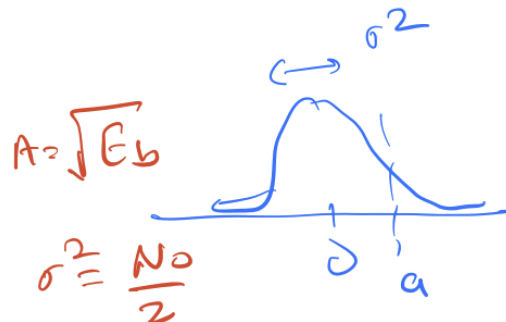
$$\begin{aligned}
 & P(\text{bit error}) \quad (y_i = -A + v) \\
 & = P(y_i > 0 \mid H_0) \text{ or } P(y_i < 0 \mid H_1) \\
 & = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(y_i - -A)^2}{\sigma^2}} = P(y_i \mid -A)
 \end{aligned}$$

$$\begin{aligned}
 & P(-A + v > 0) \\
 & : P(v > A) = P\left(\frac{v}{\sqrt{\sigma^2}} > \frac{A}{\sqrt{\sigma^2}}\right)
 \end{aligned}$$

$$\frac{v}{\sqrt{\sigma^2}} \sim N(0, 1)$$

$$\begin{aligned}
 & Q \text{ function} = P(\text{st-normal dist} > a) \\
 & \quad \underline{Q(a)}
 \end{aligned}$$

$$\begin{aligned}
 & P(v > A) \\
 & = Q\left(\frac{A}{\sqrt{\sigma^2}}\right) \\
 & = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \quad \text{where } \frac{E_b}{N_0} = \text{SNR}
 \end{aligned}$$



Symbol Error Rate (SER)

16 QAM = $2^4 = 4$ ~~8~~ bits/symbol

bit error = P_b for each bit

what is symbol error rate?

SER

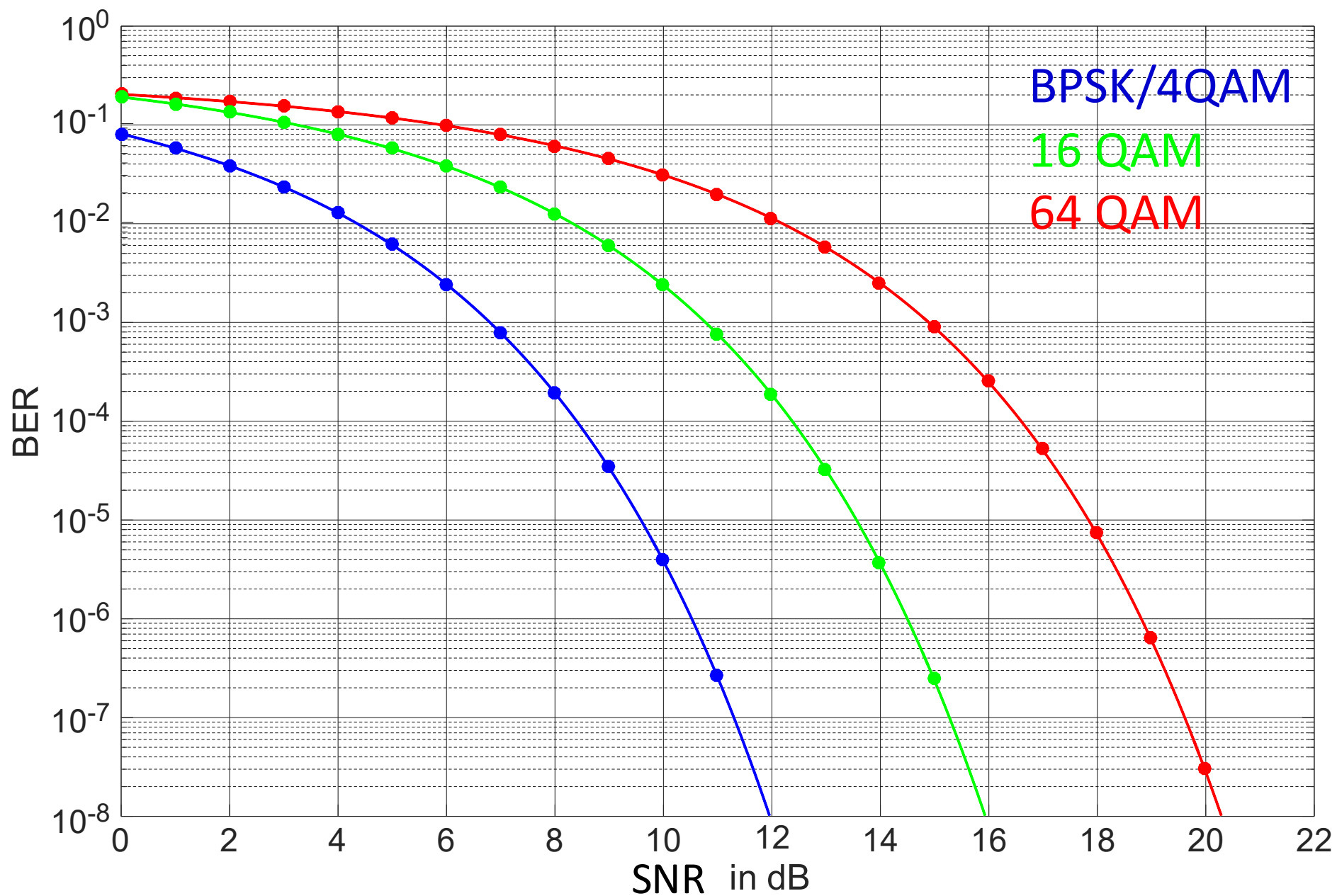
P_s = Prob that ~~any~~ one bit in error
at least

1 - no bits in error

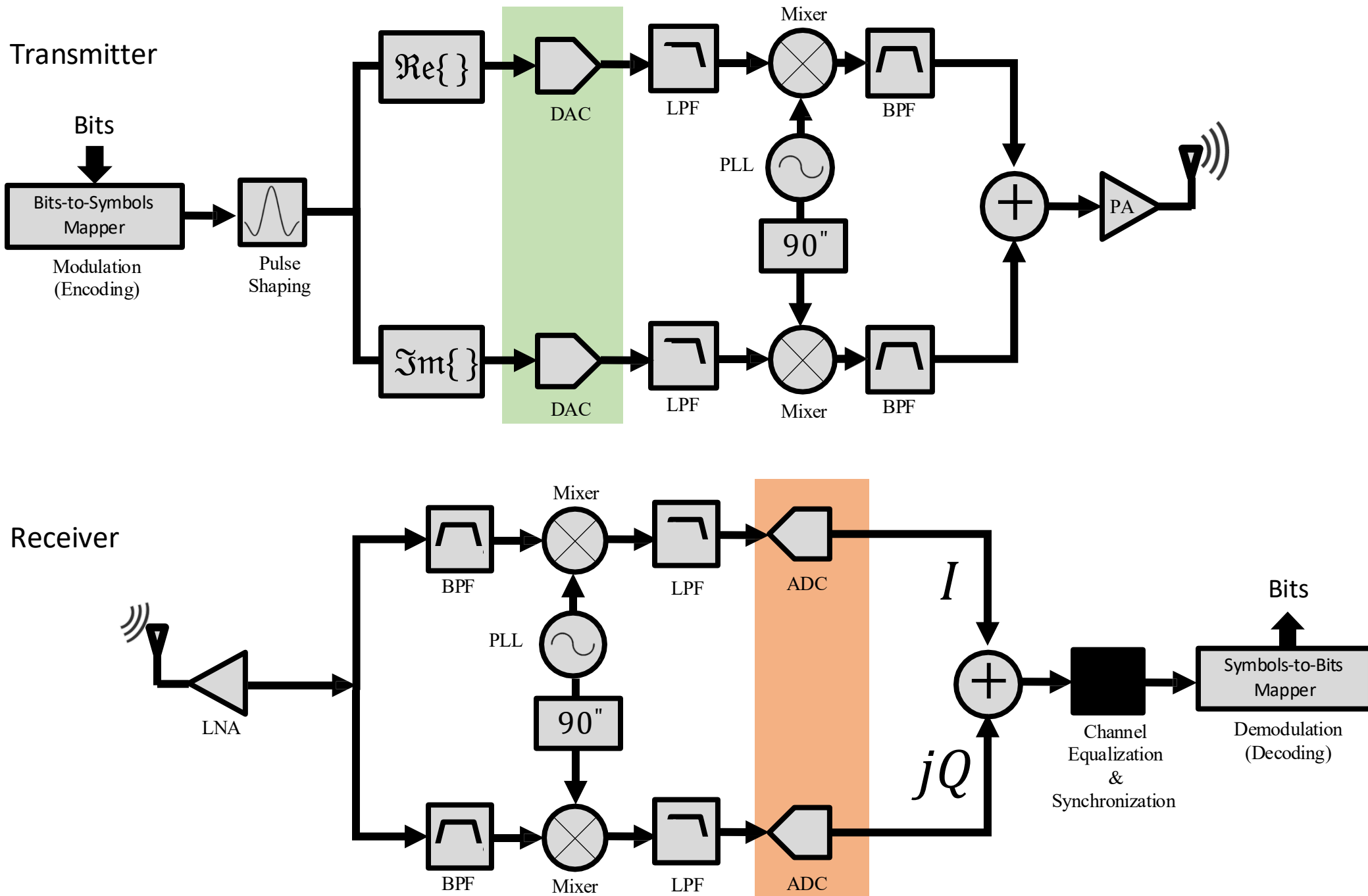
= 1 - bit 1 no error $(1 - P_b)$
x bit 2 no error $(1 - P_b)$

= $1 - (1 - P_b)^k$: k^{th} order mod.

Bit Error Rate (BER)



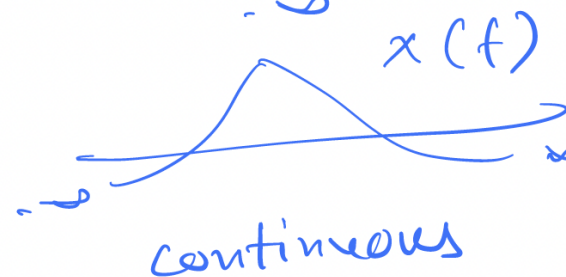
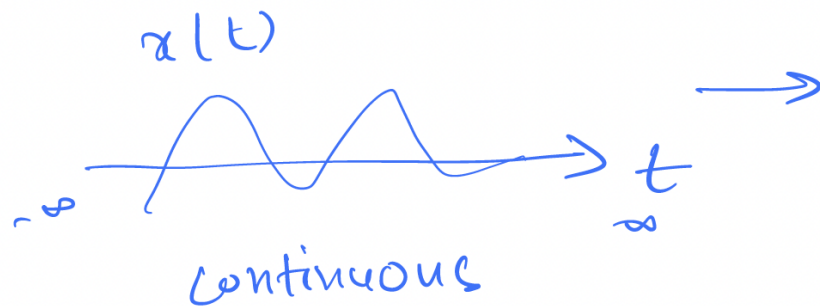
Wireless Communication System



Review: DTFT, DFT

Fourier transform
 $x(t)$

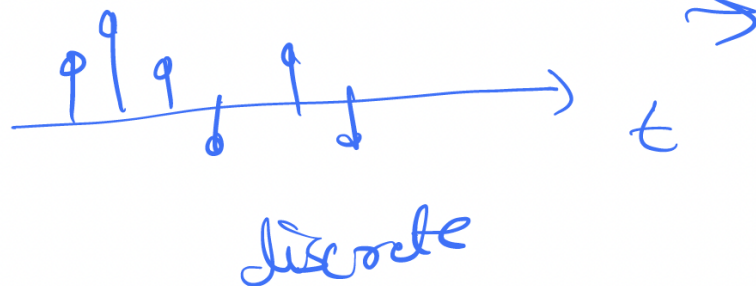
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$



DTFT

$t: nTs$

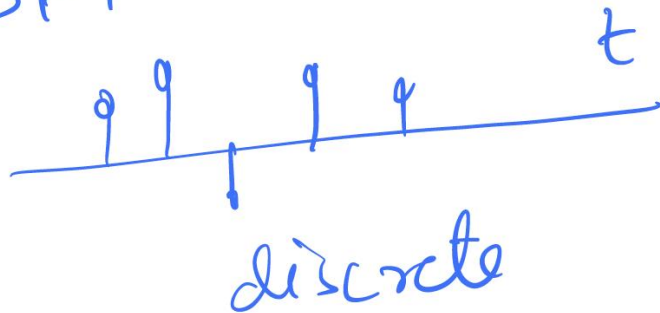
$$X(f) = \sum x(nTs) e^{-j2\pi f nTs}$$



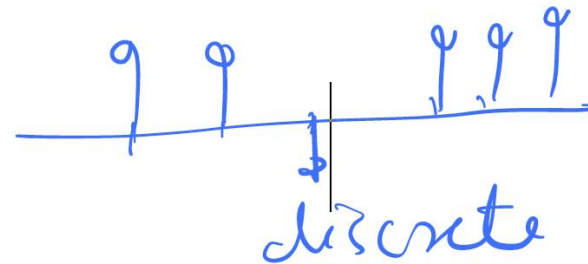
Review: DTFT, DFT

Review: DTFT, DFT

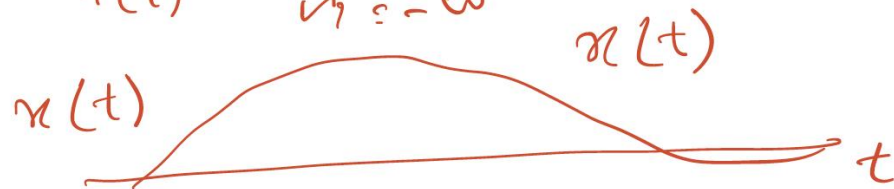
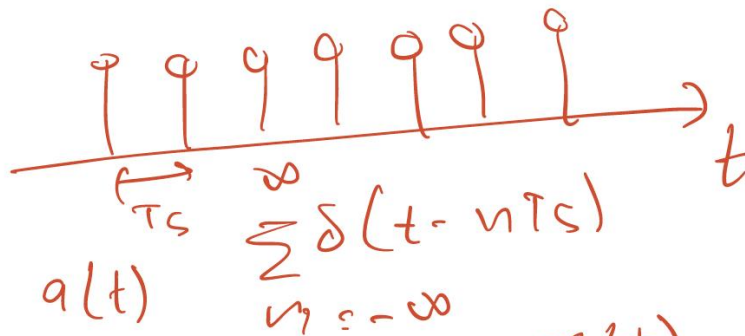
DFT



f



Pulse train



$$x_\delta(t) = a(t) \times x(t)$$

$$= x(t)$$

$$\times \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Ideal Sampling

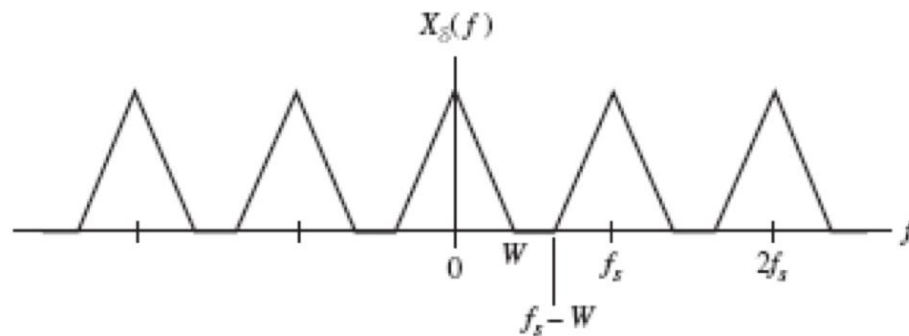
Ideal sampling

Sampling using “delta” functions

$$x_{\delta}(t) == \sum_{k=-\infty}^{\infty} x(kT_s)\delta(t - kT_s)$$
$$= x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

Spectrum

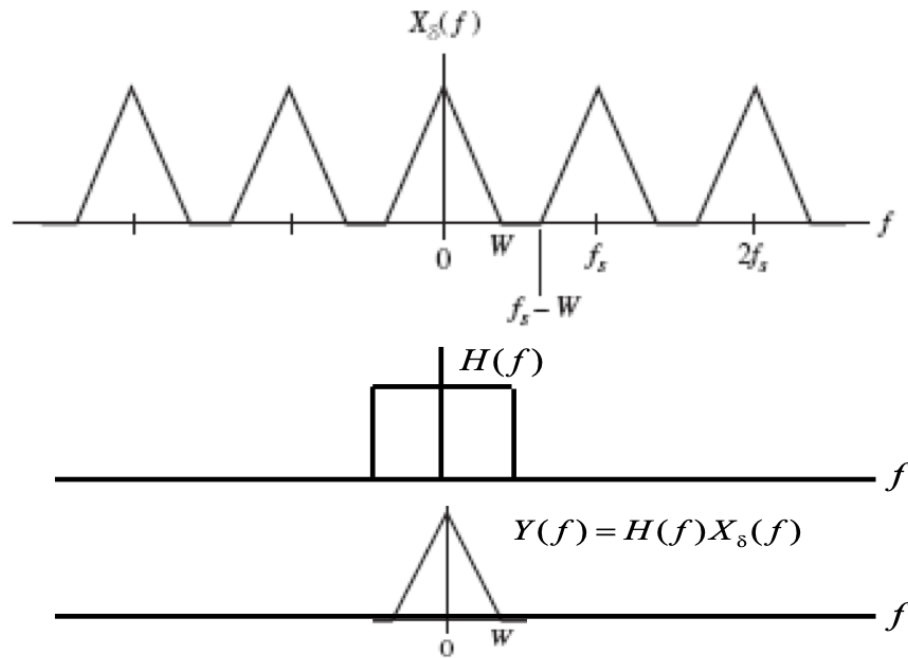
$$X_{\delta}(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$



Ideal reconstruction (1/3)

Use an LPF

With cutoff between W and $f_s - W$



Ideal reconstruction (2/3)

Ideal LPF

$$H(f) = K\Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}$$

Reconstructed signal is like original signal, but delayed by t_d , gain K , bandwidth B

Ideal reconstruction (3/3)

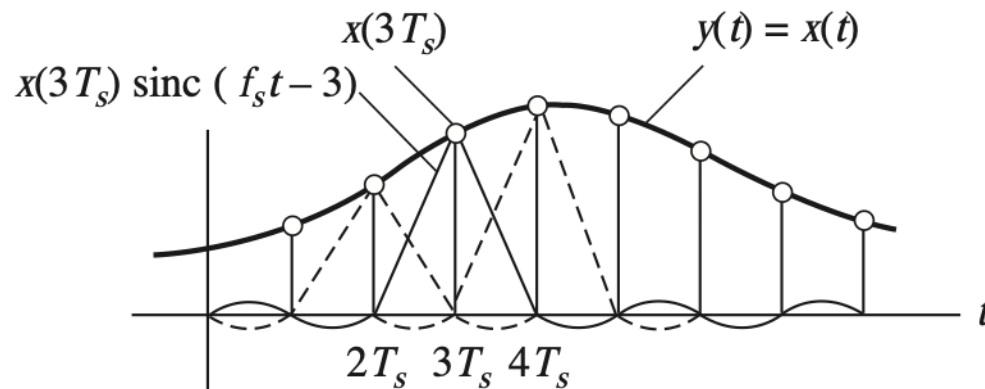
Reconstruction by interpolation (time domain)

Reconstruct from samples, but with ideal interpolation

$$h(t) = \mathcal{F}^{-1}[H(f)] = 2BK \operatorname{sinc} 2B(t - t_d)$$

$$\Rightarrow y(t) = h(t) * x(kT_s) = 2BK \sum_{k=-\infty}^{\infty} x(kT_s) \operatorname{sinc} 2B(t - t_d - kT_s)$$

(Unimplementable in practice)



Practical sampling & reconstruction

Real samplers

- Have finite duration pulses (not delta functions)

Real reconstruction filters

- Not ideal (or distortion-less filters)
- Must be causal (in most cases)

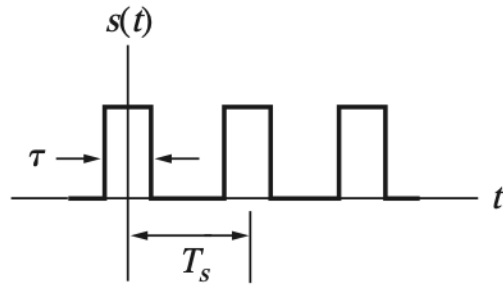
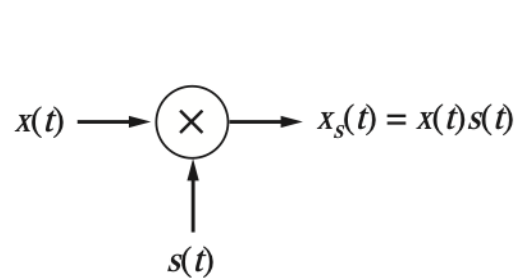
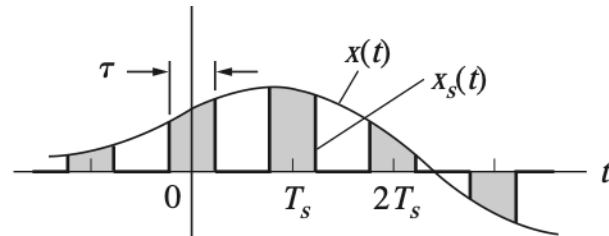
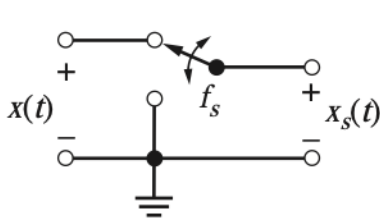
Real signals

- Are time-limited, and therefore cannot be band-limited
- $\Rightarrow$ Signal *distortion* and *aliasing* cannot be entirely avoided when using sampling and reconstruction

Non-ideal sampling (chopper sampling)

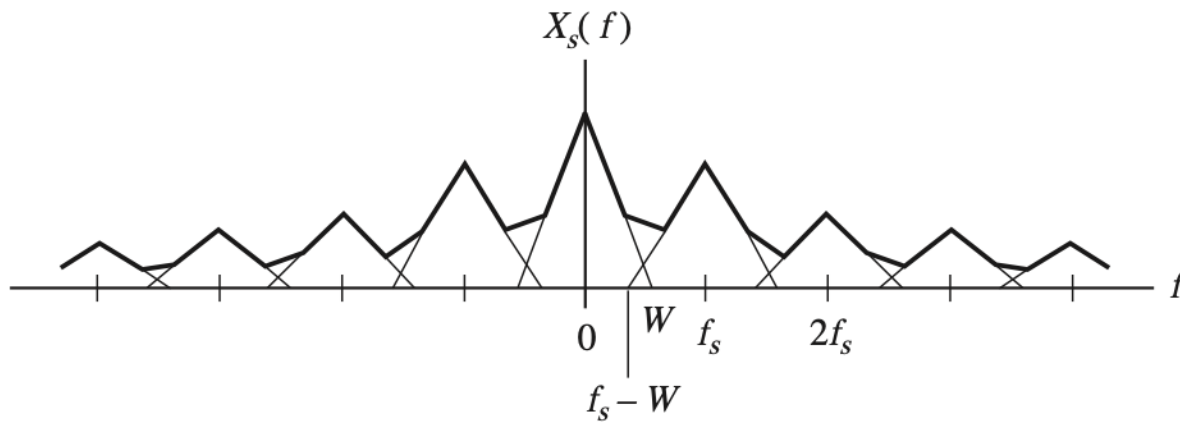
Pulse sampling or chopper sampling

$x_s(t)$: A sampled version of $x(t)$



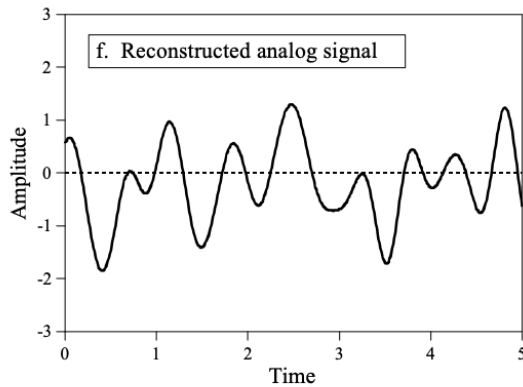
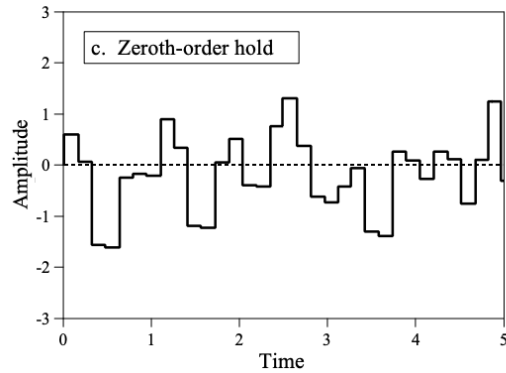
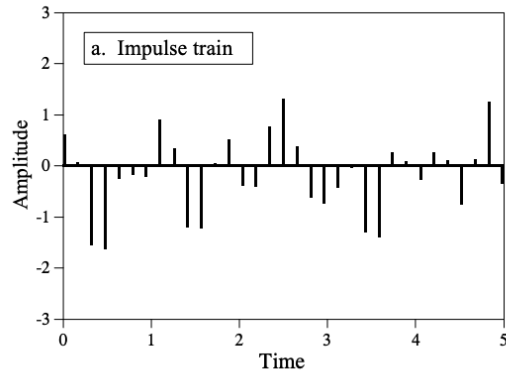
Aliasing

Under-sampled message

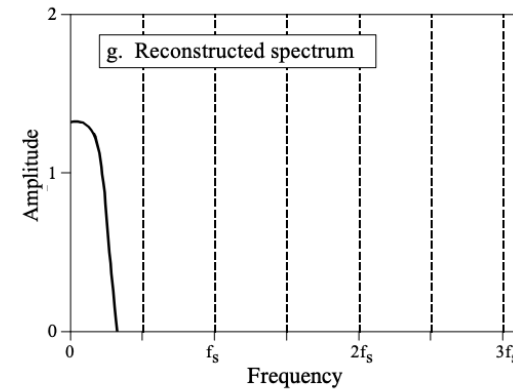
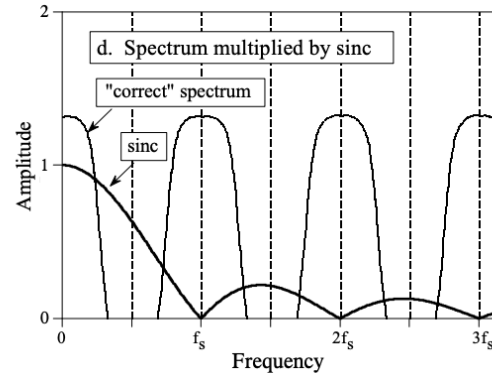
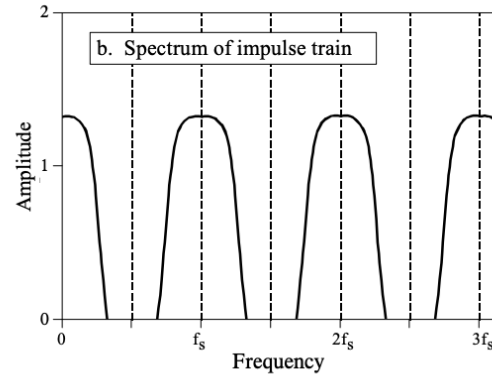


DAC Design and reconstruction

Time Domain

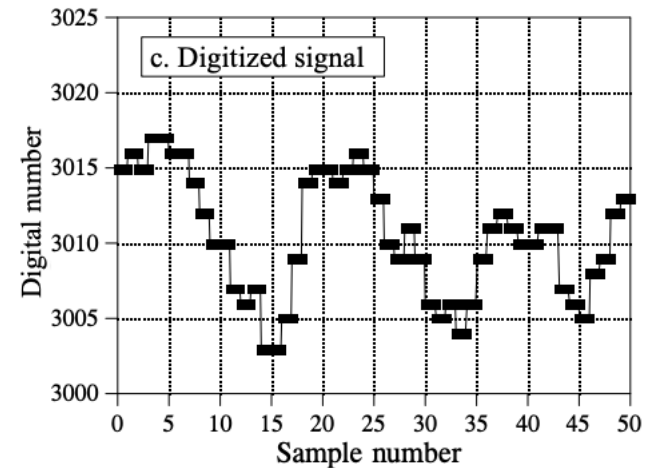
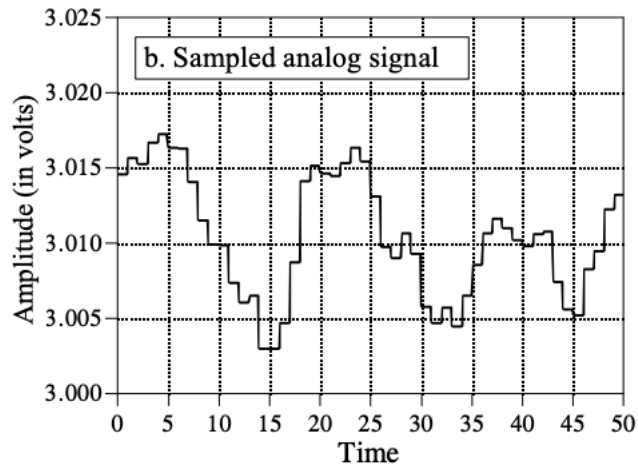
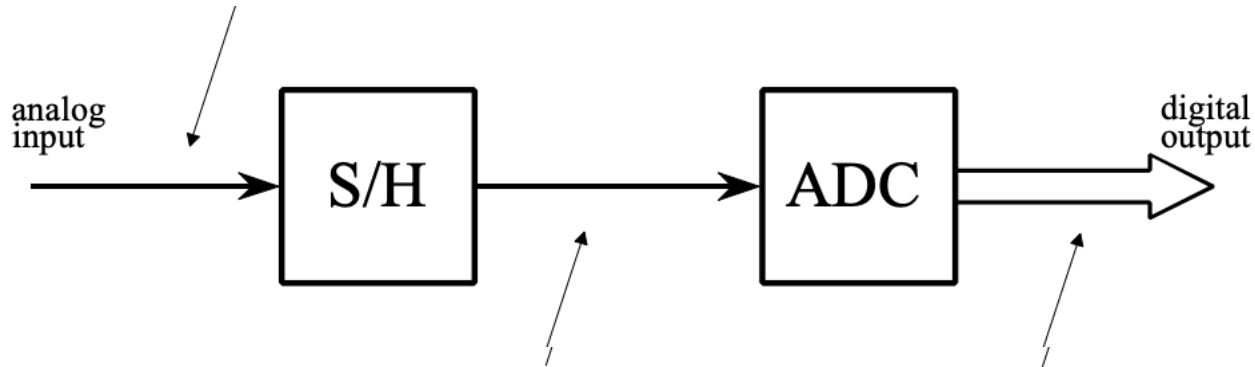
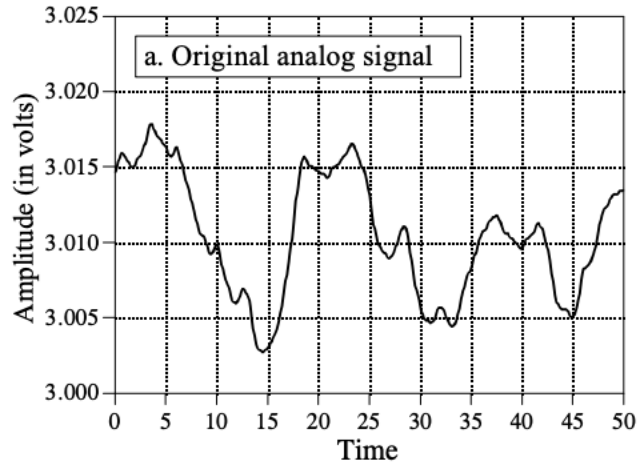


Frequency Domain



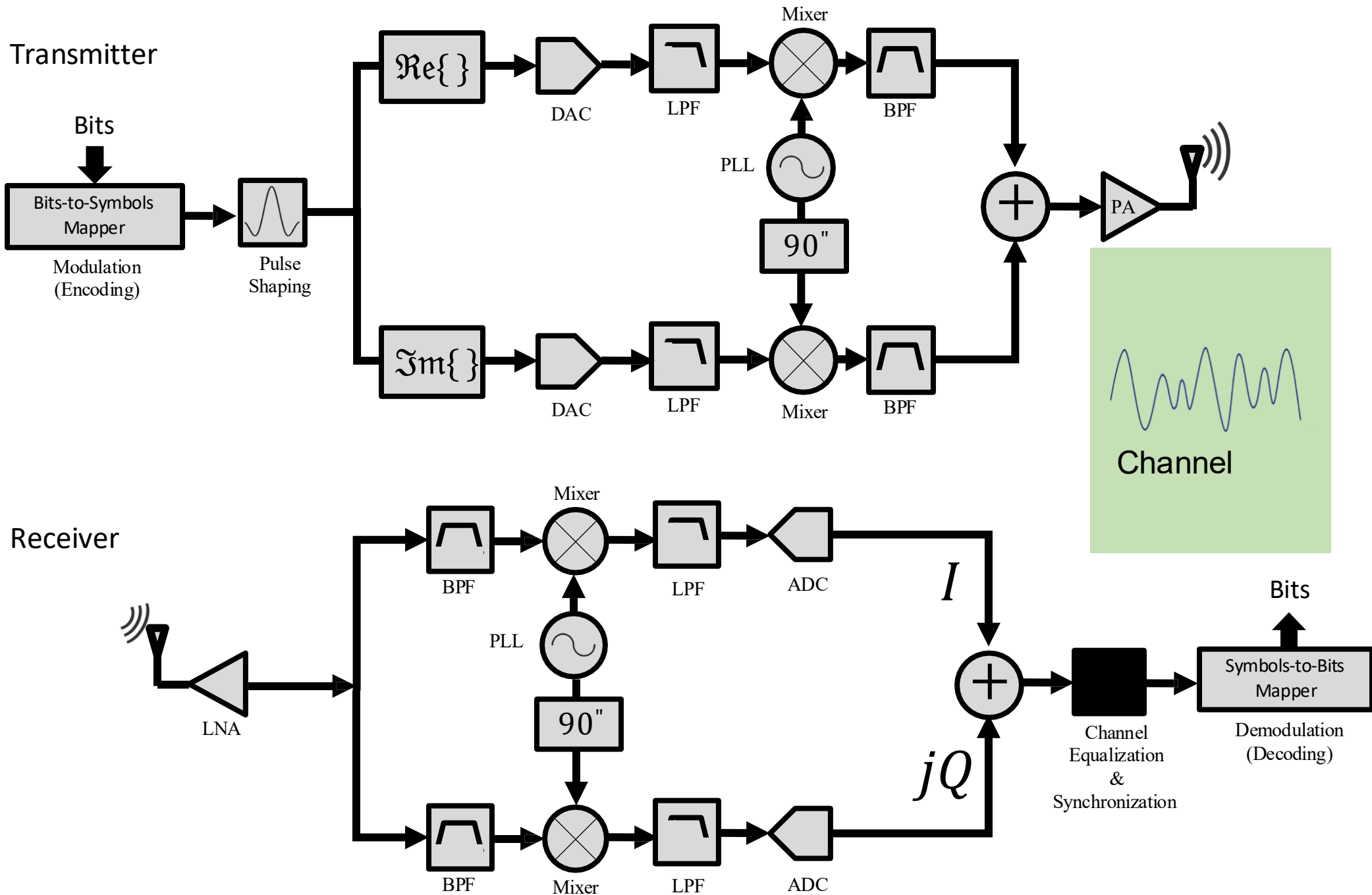
DAC Design and reconstruction

ADC Design with Quantization

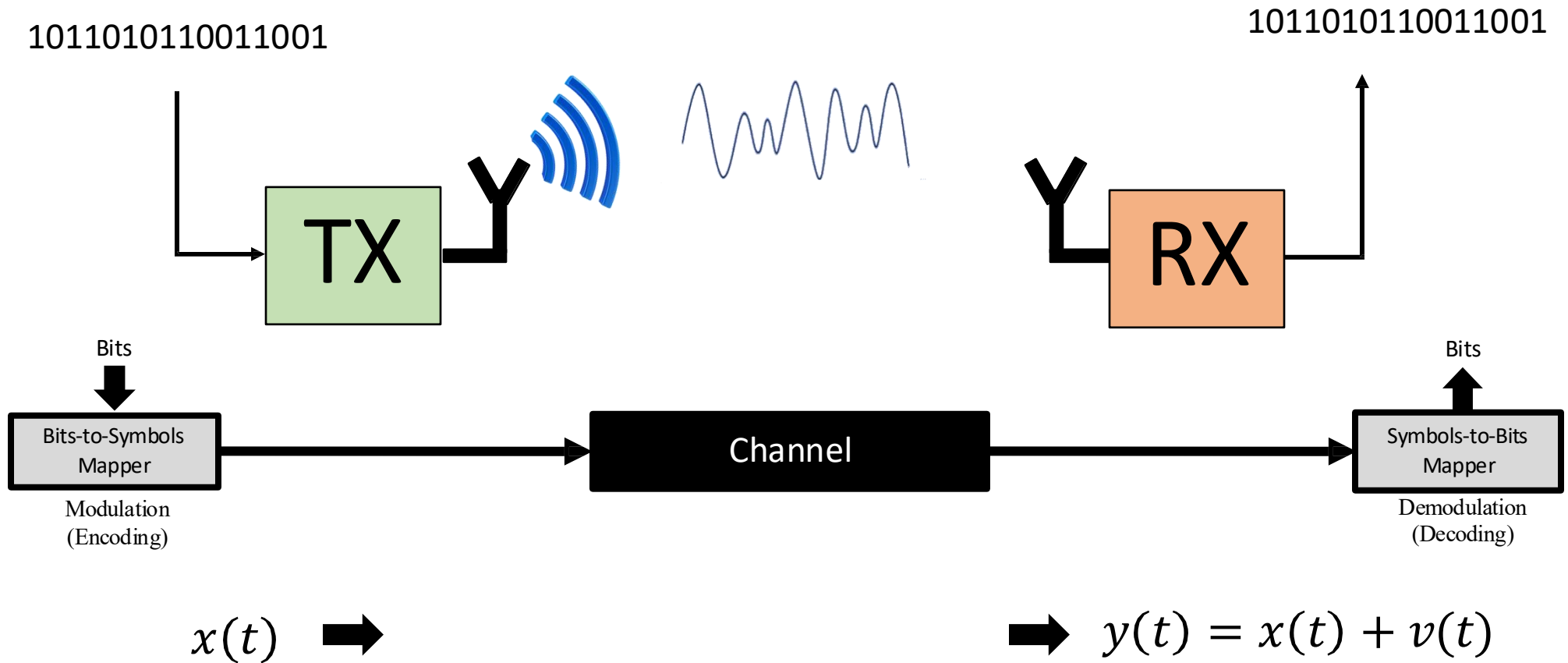


ADC Design with Quantization

Wireless Communication System



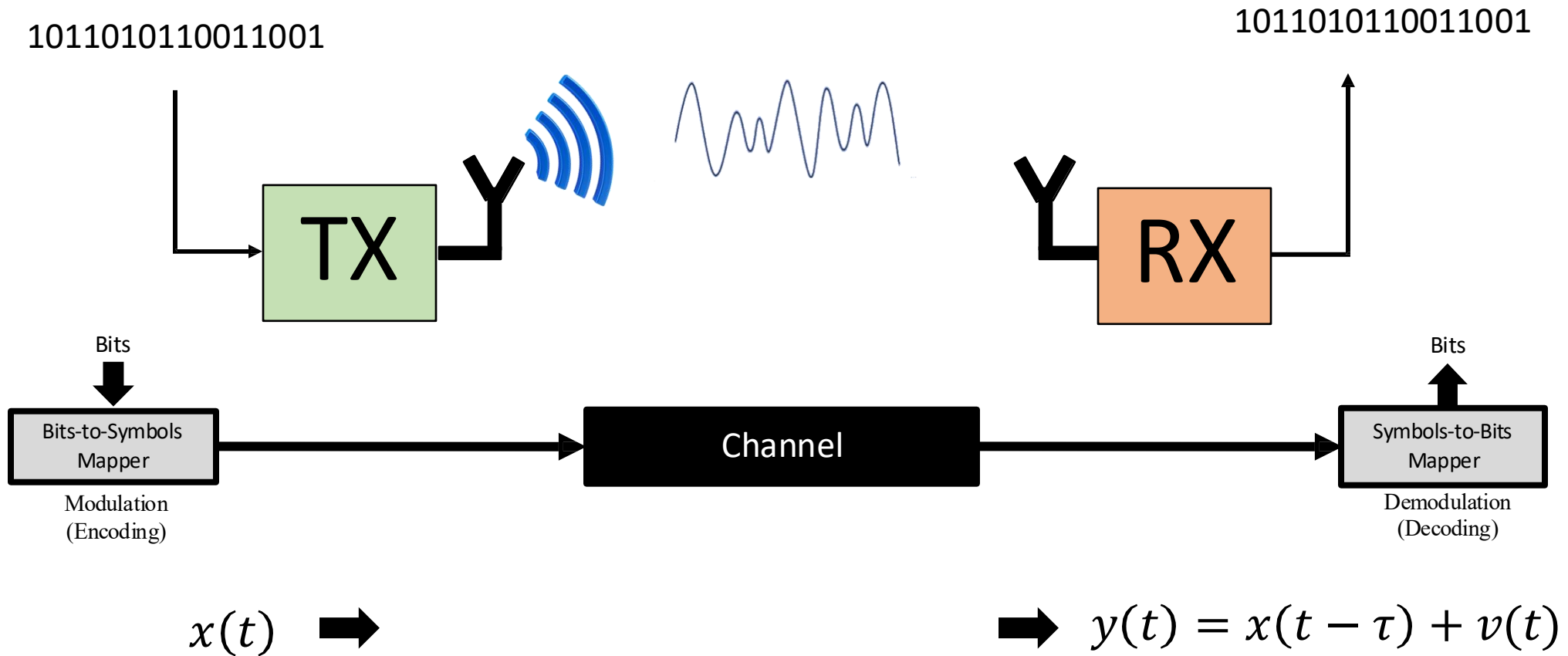
The Channel



Channel adds noise (AWGN)!

$$v(t) \sim N(0, \sigma^2)$$

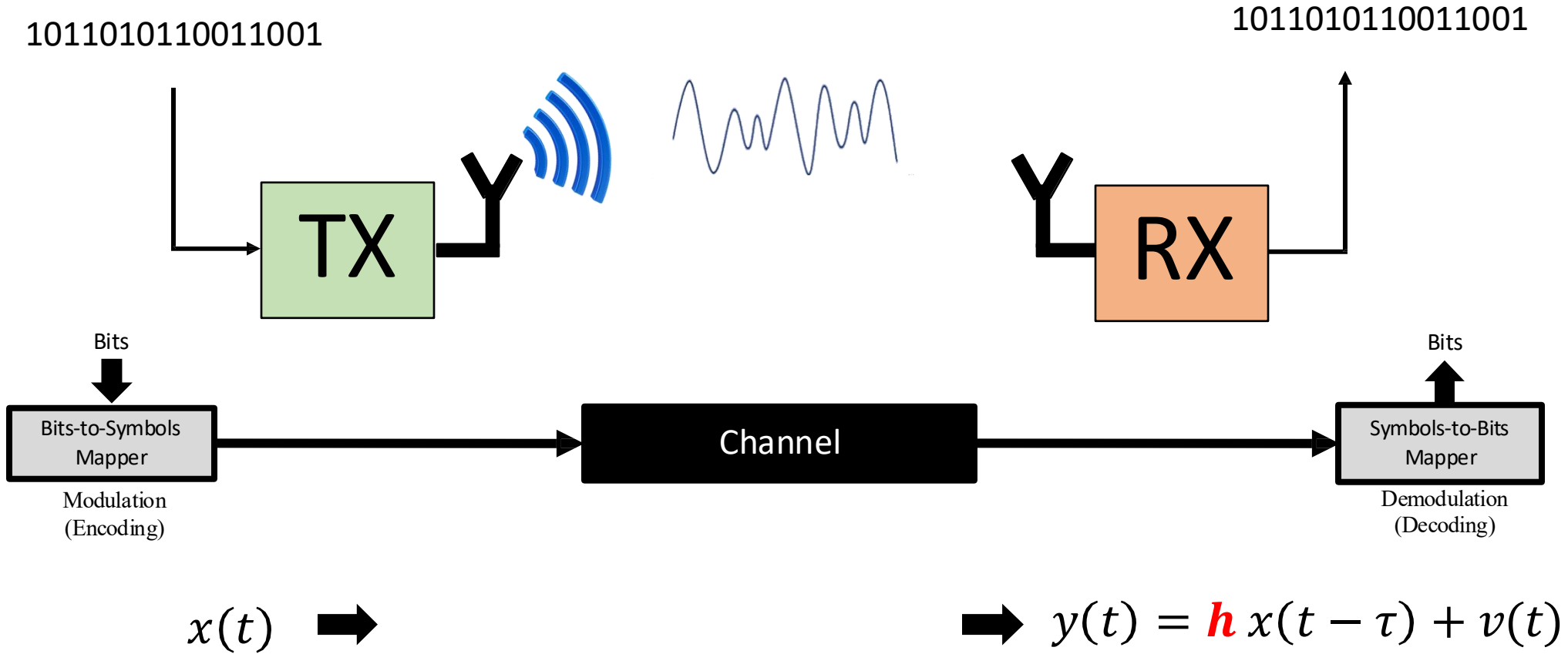
The Channel



Channel delays the signal!

$$\tau = \frac{d}{c}$$

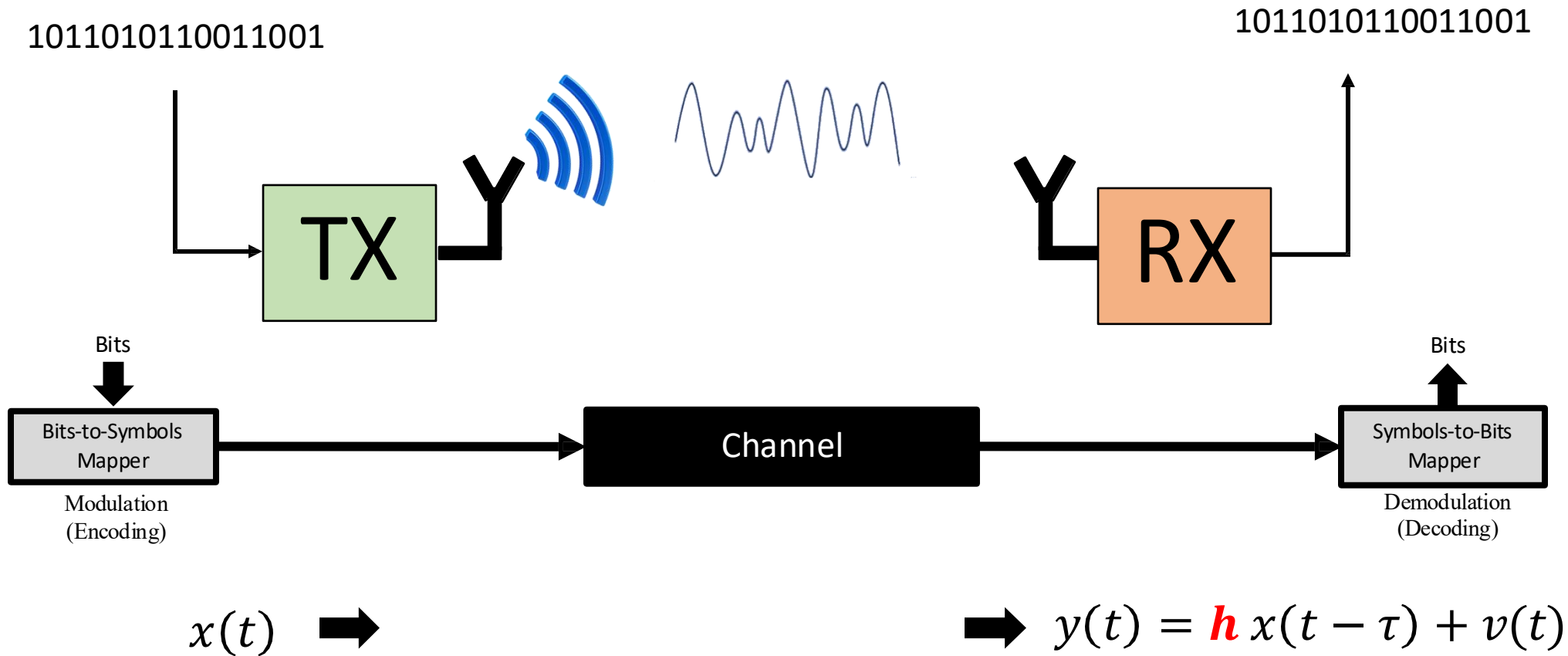
The Channel



Channel attenuates the signal (Pathloss)

$$P_{\text{RX}} = G_{\text{TX}} G_{\text{RX}} \frac{\lambda^2}{(4\pi d)^2} P_{\text{TX}} \quad \Rightarrow \quad |h| \propto \frac{\lambda}{d}$$

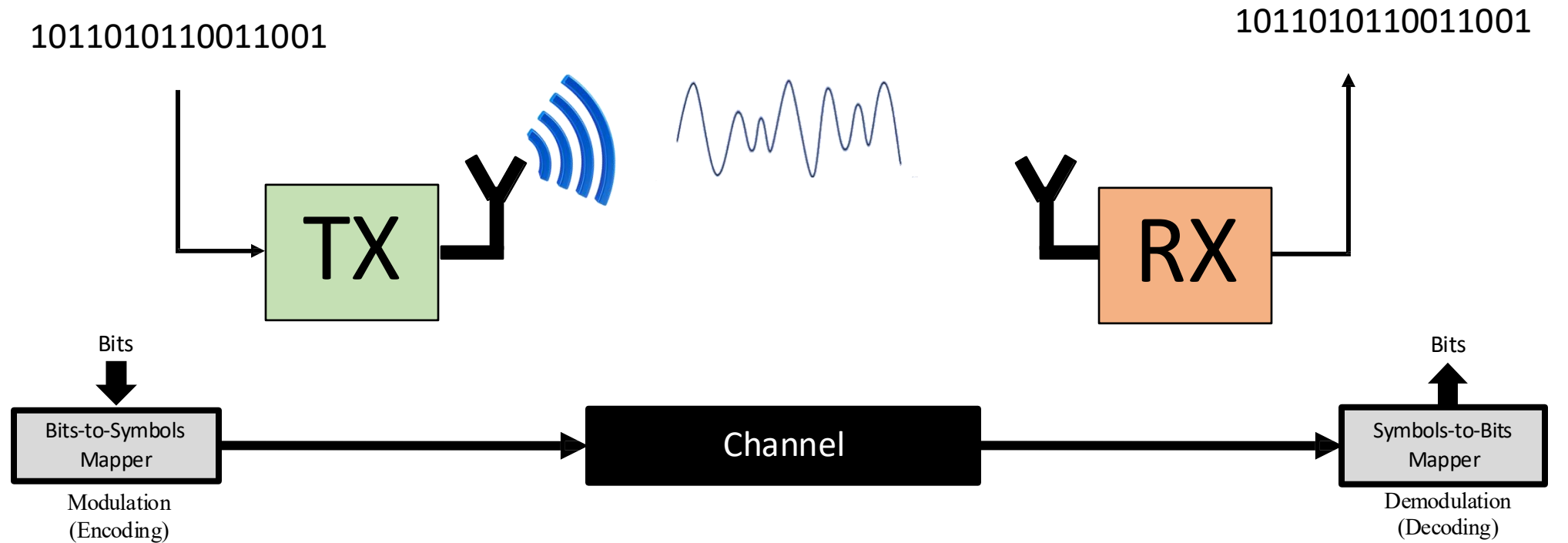
The Channel



Channel rotates the signal (Adds Phase)

$$h \propto \frac{\lambda}{d} e^{j\phi}$$

The Channel



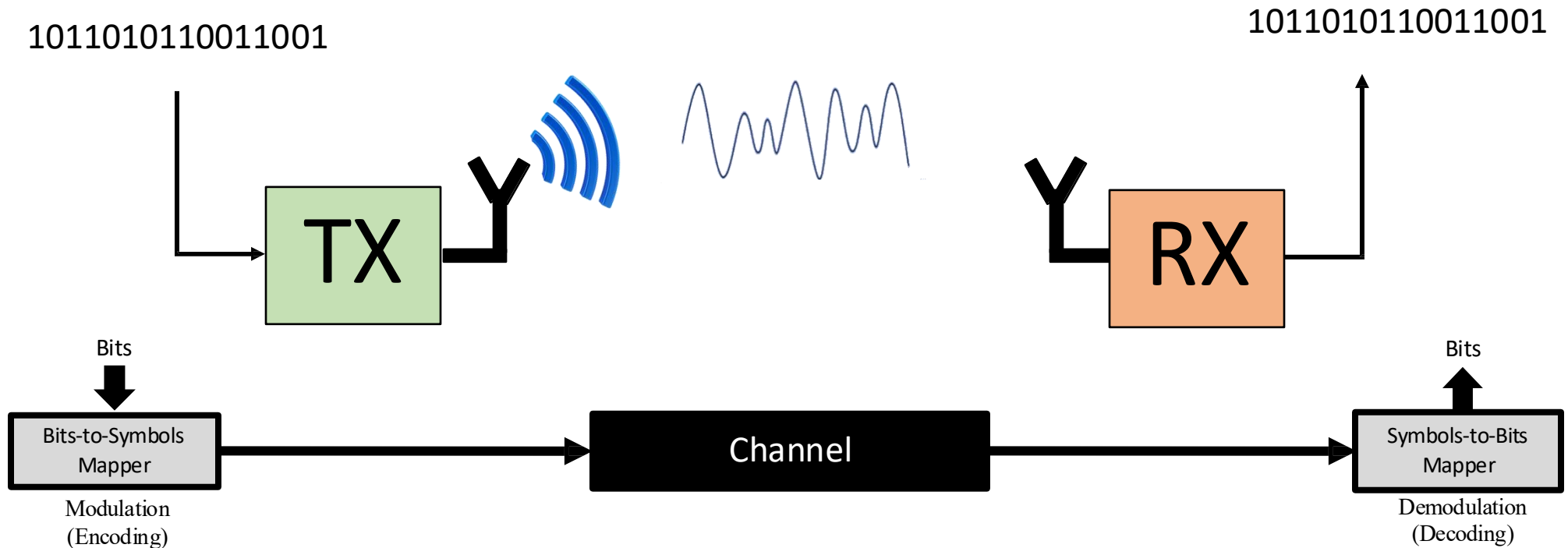
$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

$$x(t) \times e^{-j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{-j2\pi f_c (t - \tau)} \Rightarrow \times e^{j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{j2\pi f_c t}$$

$$\mathbf{h} \propto \frac{\lambda}{d} e^{j\phi} \rightarrow \phi = 2\pi f_c \tau = 2\pi \frac{c}{\lambda} \frac{d}{c} = 2\pi \frac{d}{\lambda} \rightarrow \mathbf{h} \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel



$$x(t) \rightarrow$$

$$\rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

Channel:

- Adds Noise
- Delays the Signal
- Attenuates the Signal
- Rotates the Phase of the Signal

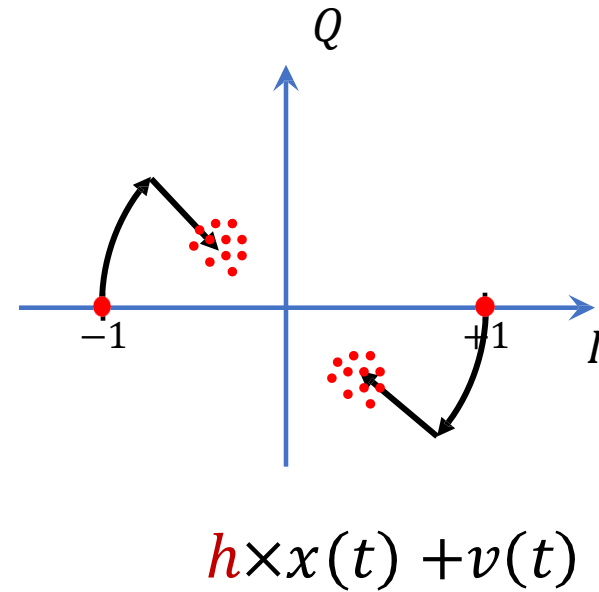
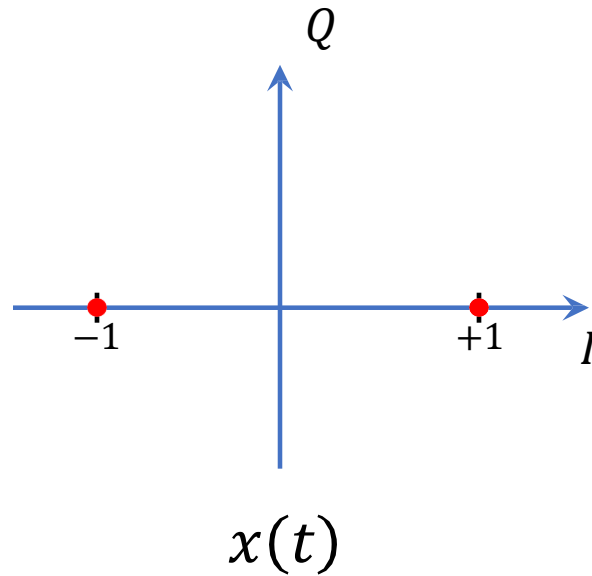
$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

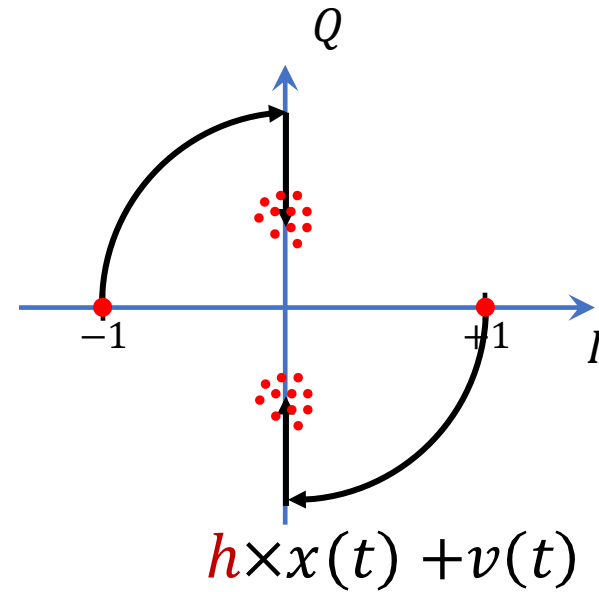
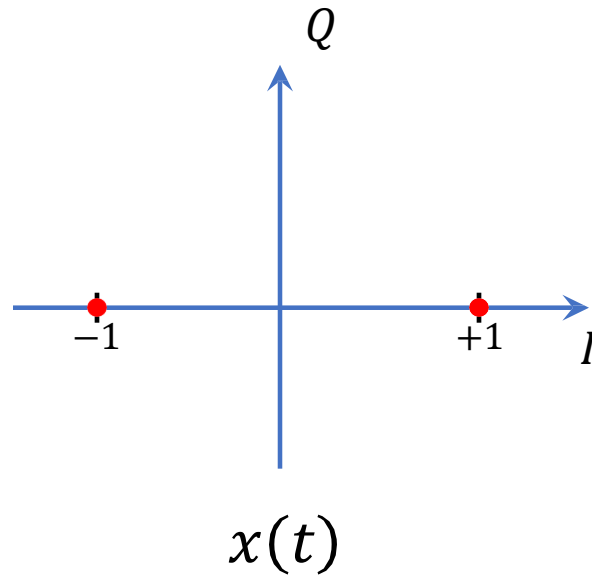


The Channel

Consider BPSK Modulation.

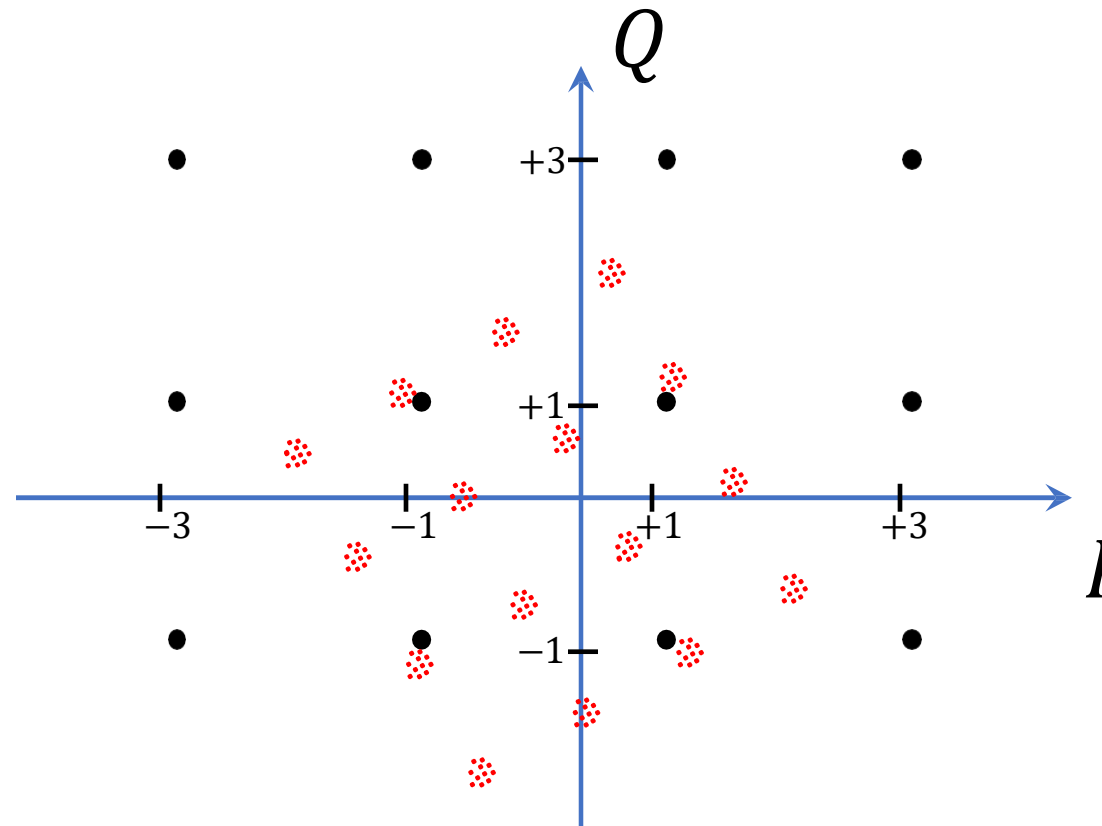
$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



The Channel

Consider QAM Modulation



Demodulating correctly requires COHERENCE!
i.e., Need to estimate & correct for the channel h



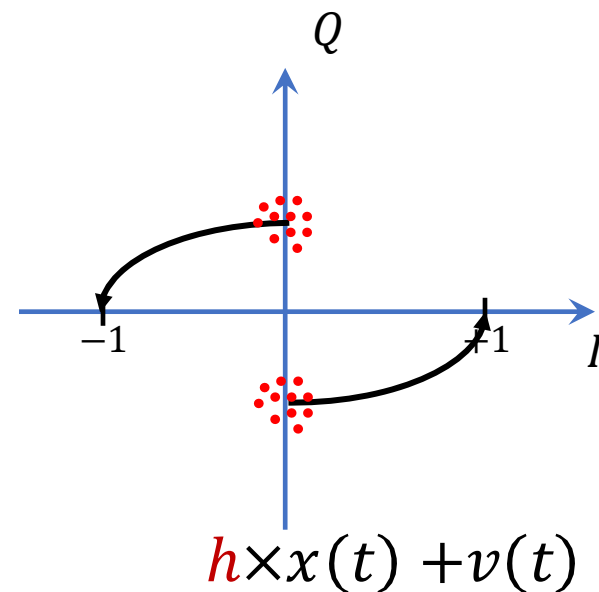
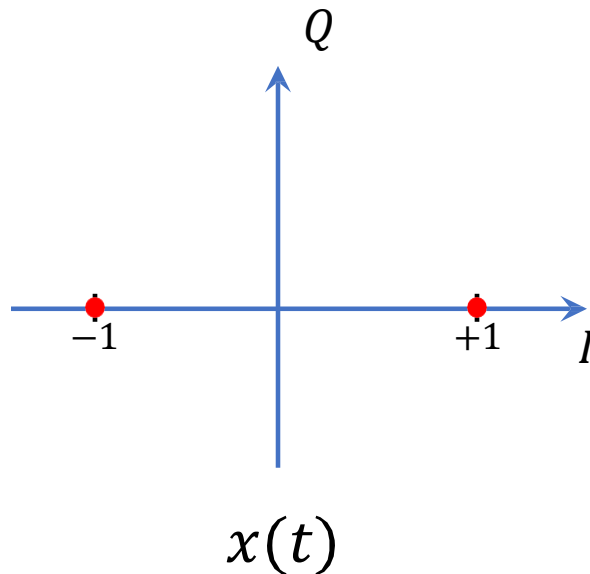
Channel Estimation & Correction

Channel Estimation & Correction

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



Send Training Sequence (Preamble Bits): Known Bits

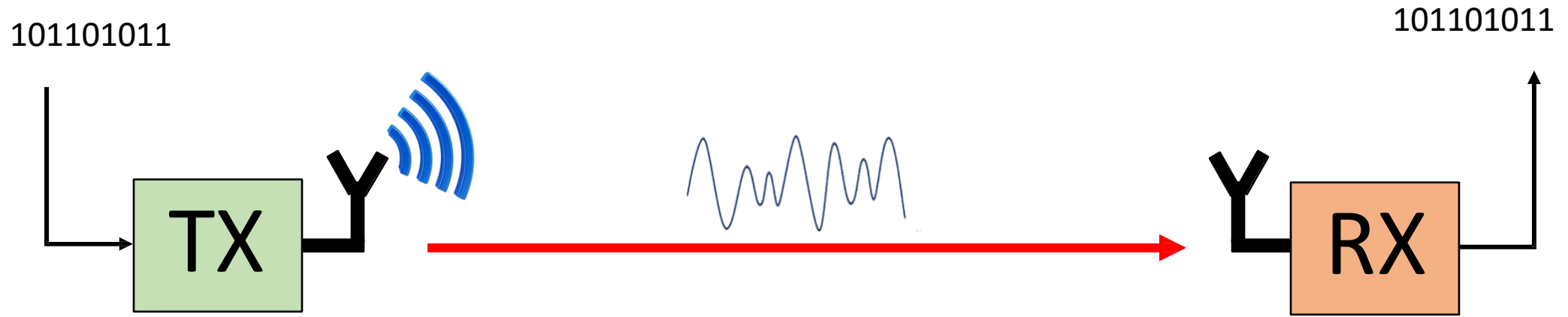
$$\begin{aligned} x(0) = 1 & \longrightarrow y(0) = h + v(0) \\ x(1) = 1 & \longrightarrow y(1) = h + v(1) \\ x(2) = -1 & \longrightarrow y(2) = -h + v(2) \end{aligned}$$

•
•
•

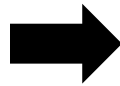
$$\text{Estimate channel: } \tilde{h} = \frac{1}{K} \sum_{k=1}^K \frac{y(k)}{x(k)}$$

$$\text{Correct channel: } \tilde{x}(t) = \frac{y(t)}{\tilde{h}(t)}$$

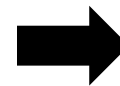
The Channel



$x(t)$



Channel

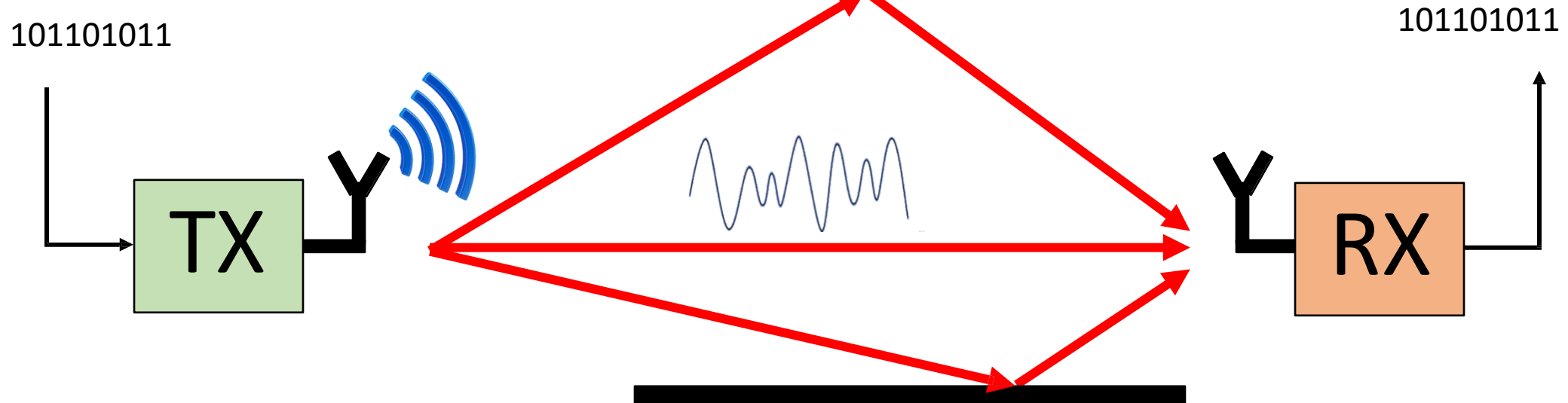


$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

Assumes single path!

Multipath Channel



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

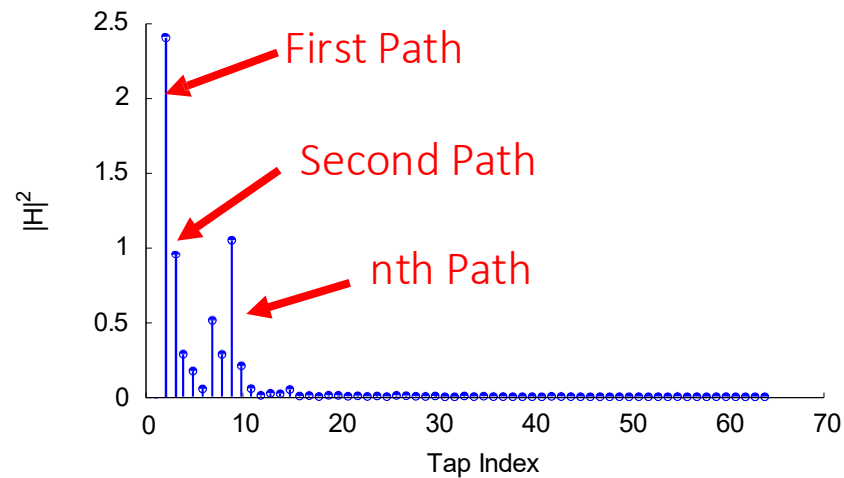
$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$h(t)$ is channel impulse response.

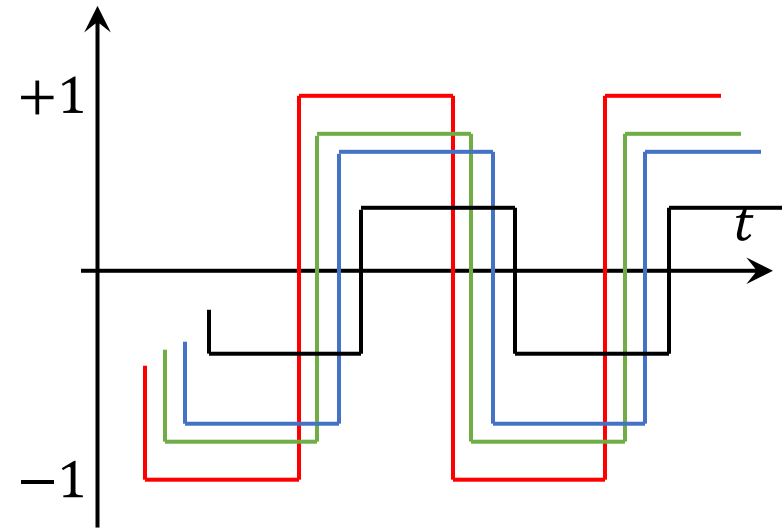
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



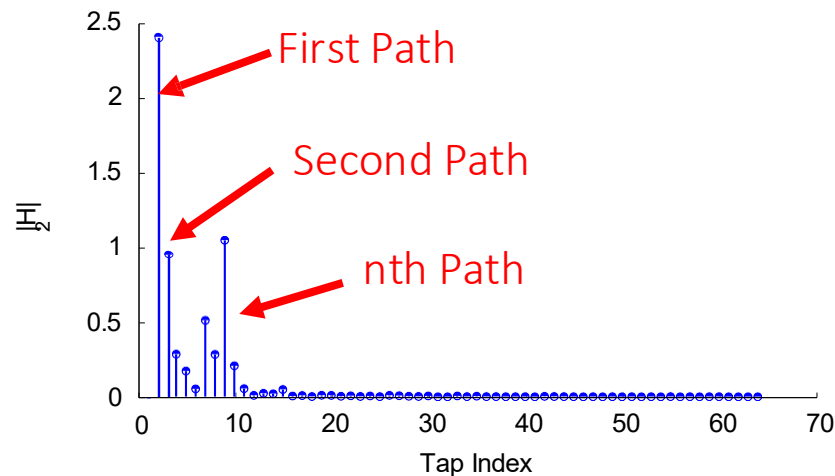
ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.

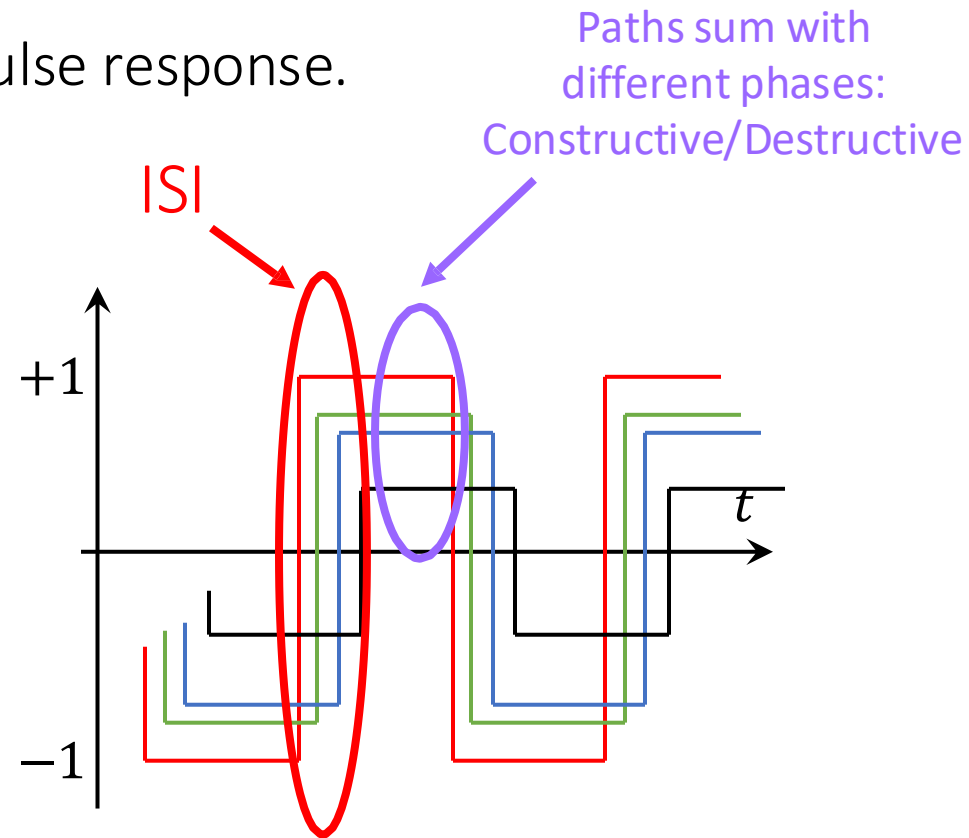
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Channel Fading

Symbols arriving along different paths sum up destructively

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi(d_2-d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1$$

$$\text{if } \frac{d_2 - d_1}{\lambda} \approx \frac{1}{2} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + e^{j\pi}) = 0 \quad \text{Destructive Interference!}$$

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

@ $f_1 = 2.5GHz$ ($\lambda = 12\text{ cm}$): $h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$

@ $f_2 = 5GHz$ ($\lambda = 6\text{ cm}$): $h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$

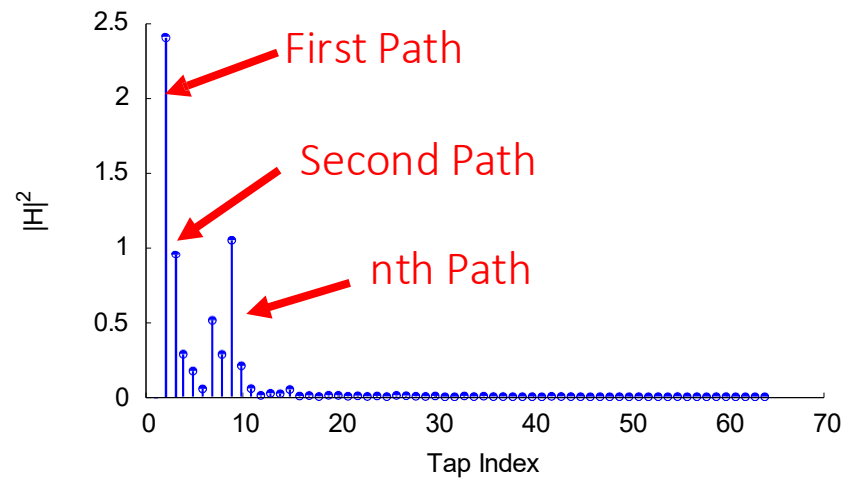
Frequency Selective Fading

17×
(24dB)

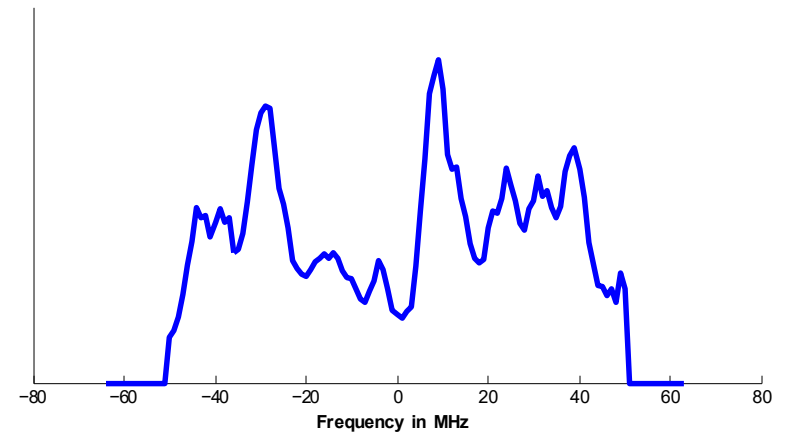
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



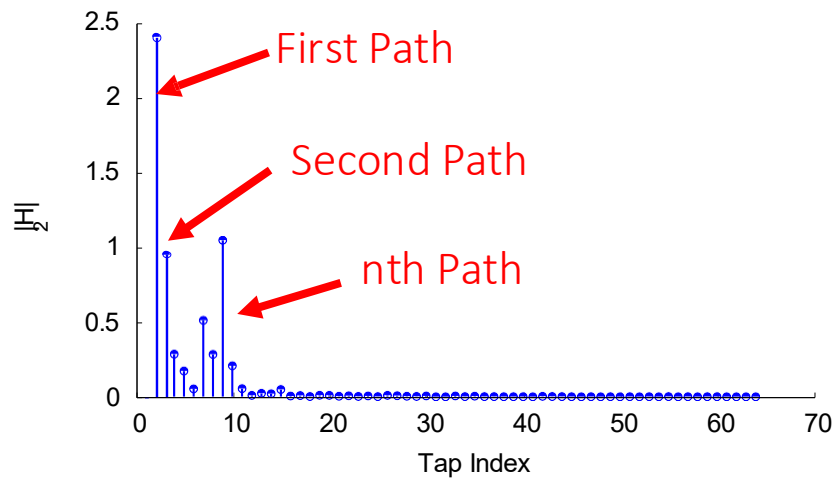
Multi-tap Channel



Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$

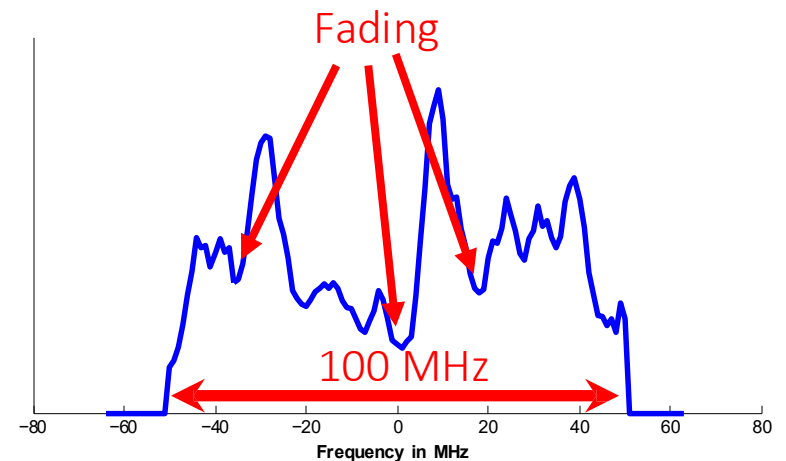


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

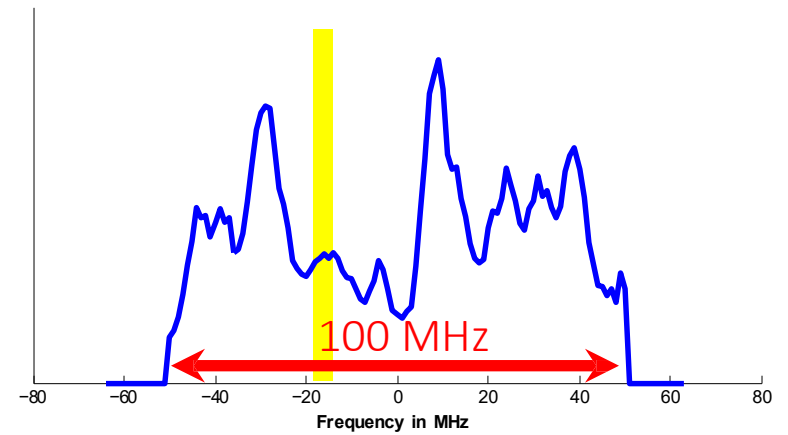
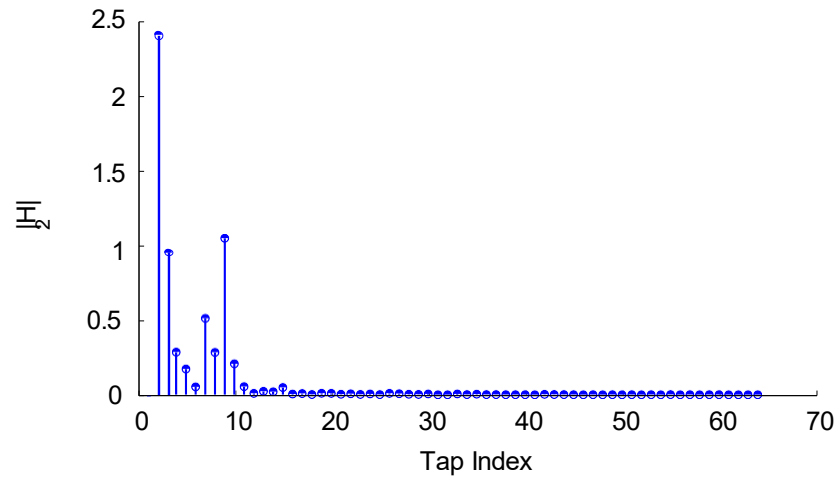
Symbols arriving along different paths sum up destructively

Problematic in
Wideband Channel!

Narrowband Channel

$h(t)$ is channel impulse response.

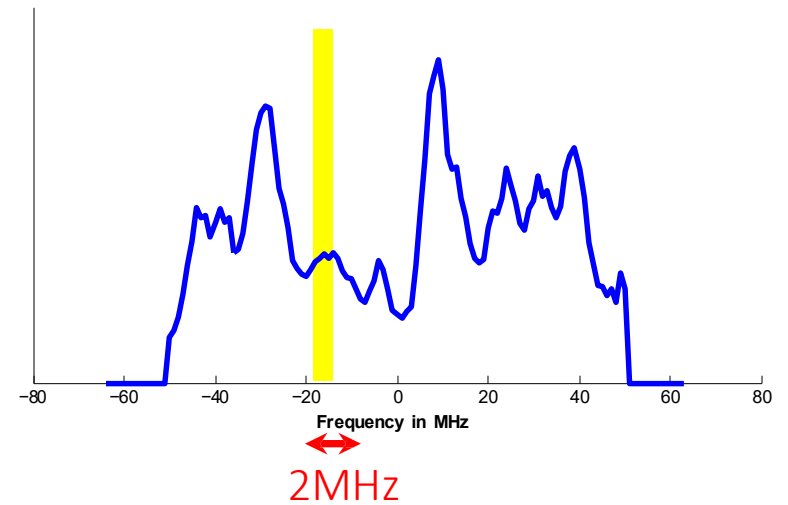
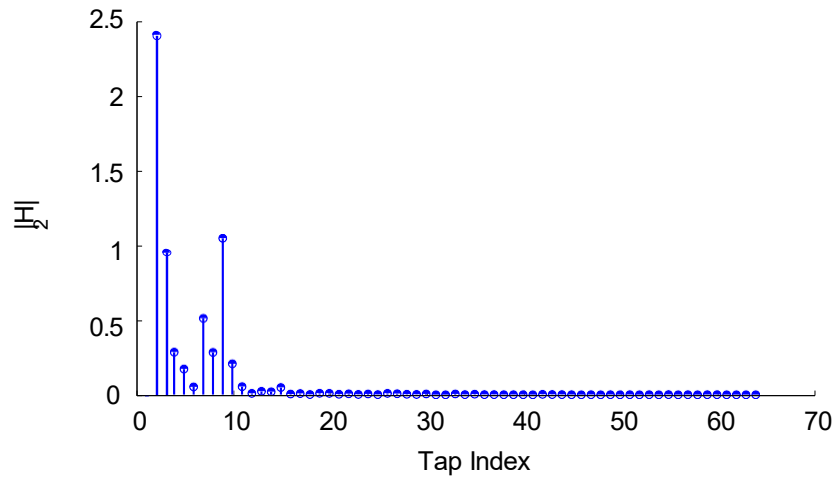
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

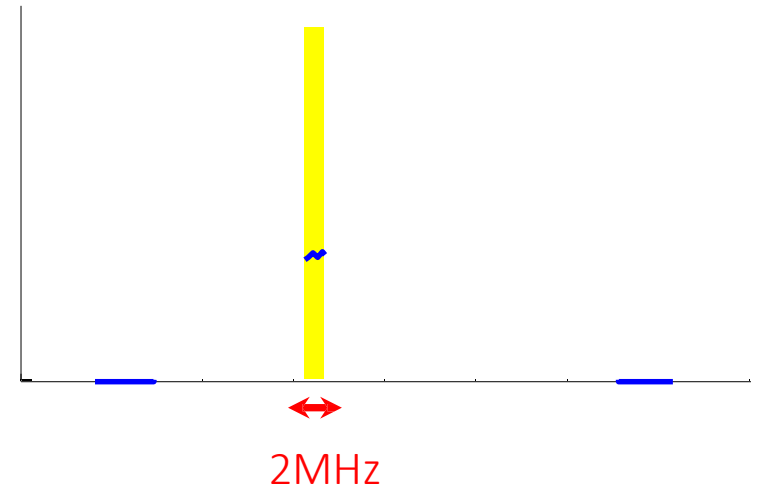
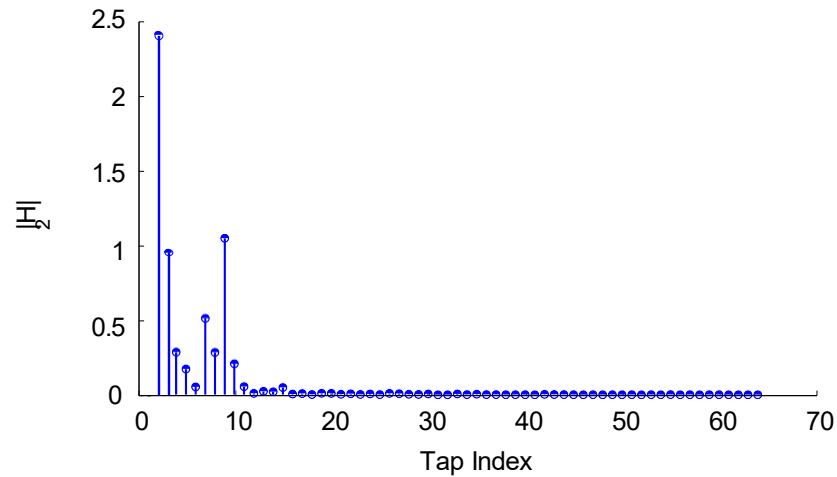
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

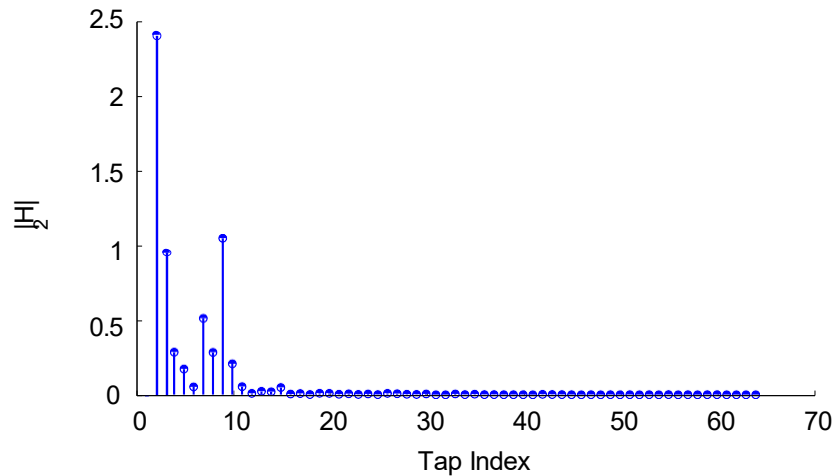
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



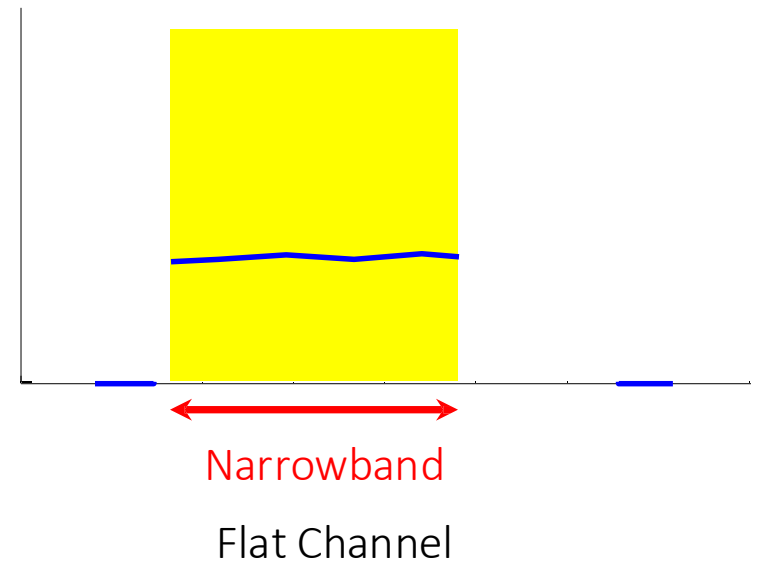
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



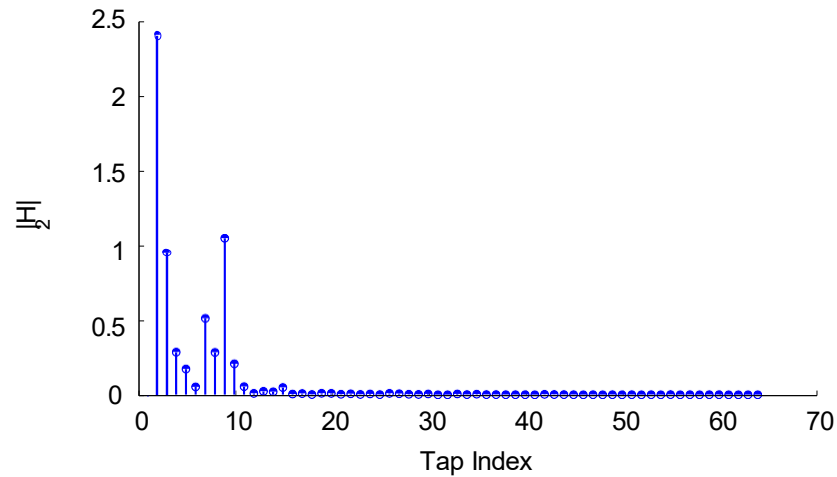
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



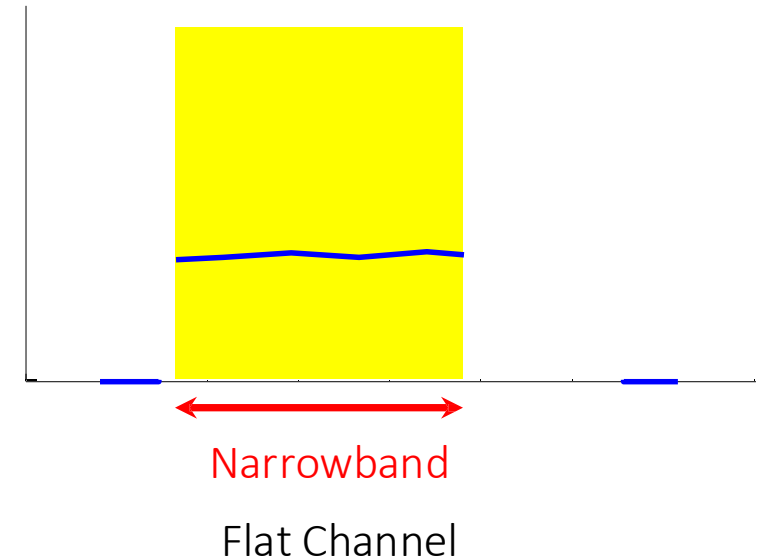
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



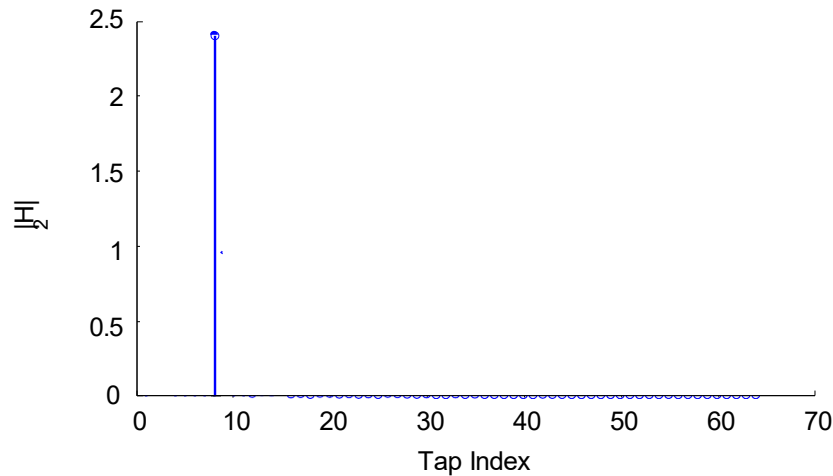
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



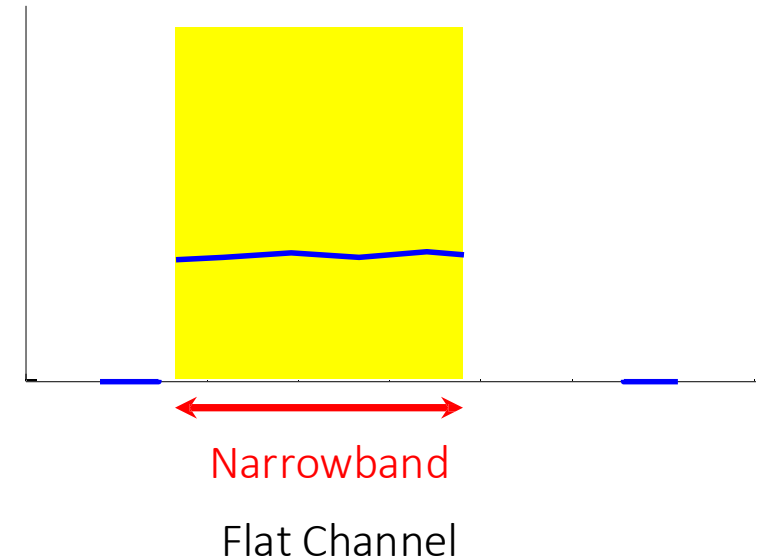
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



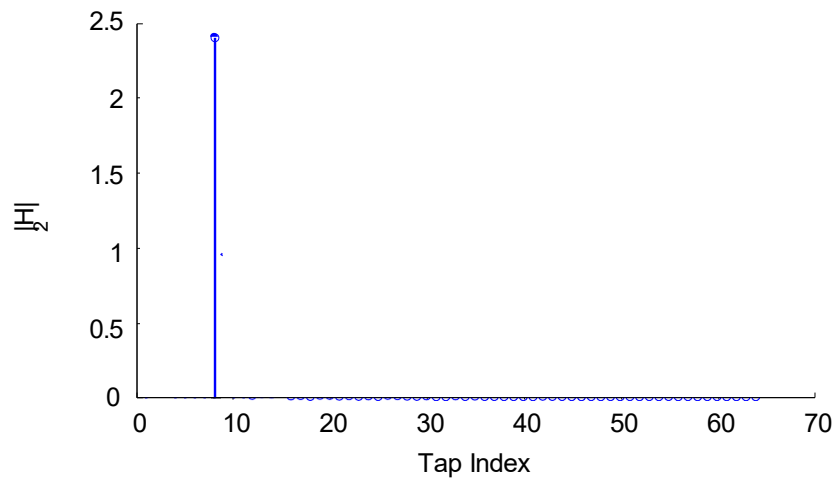
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

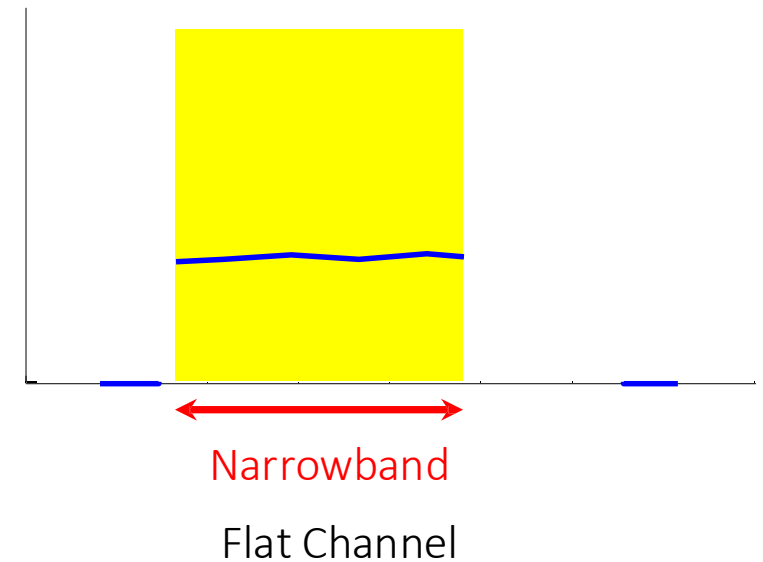
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



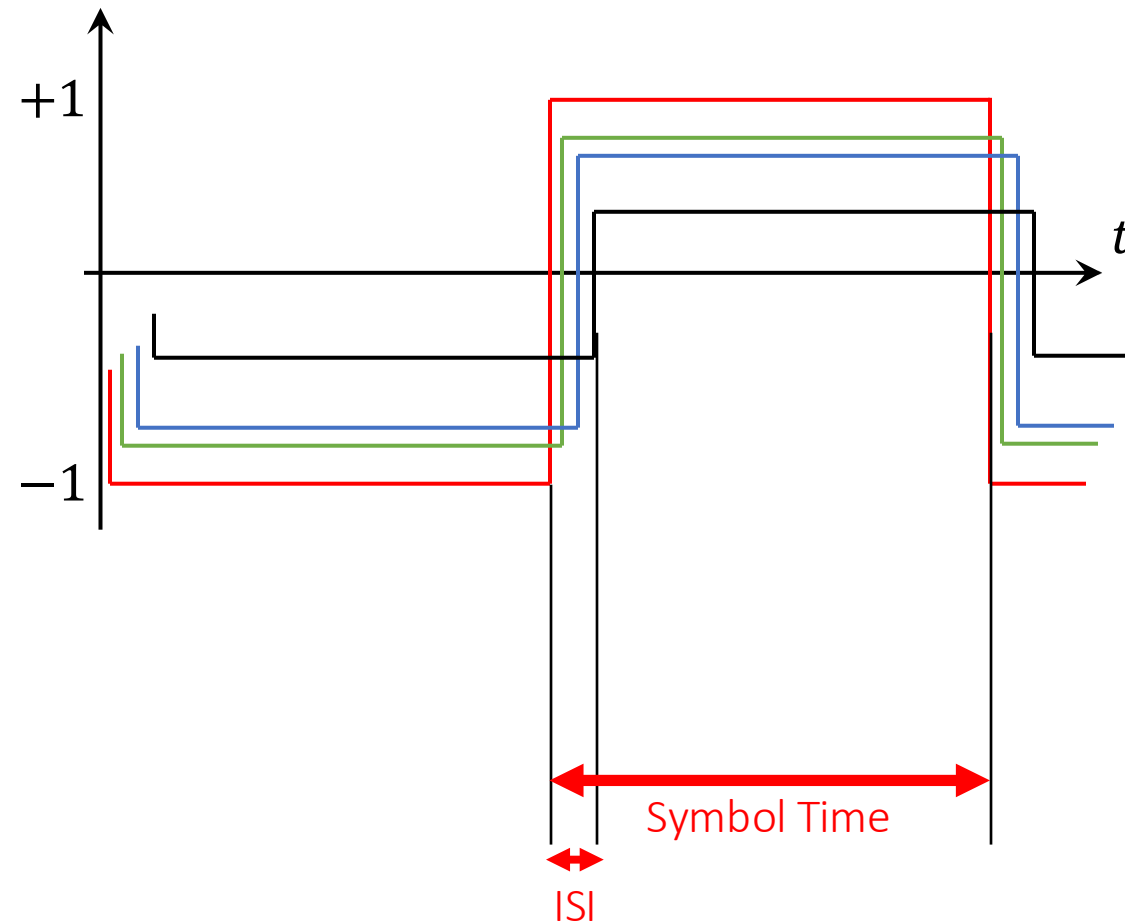
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = h x(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

Narrowband vs. Wideband Channel

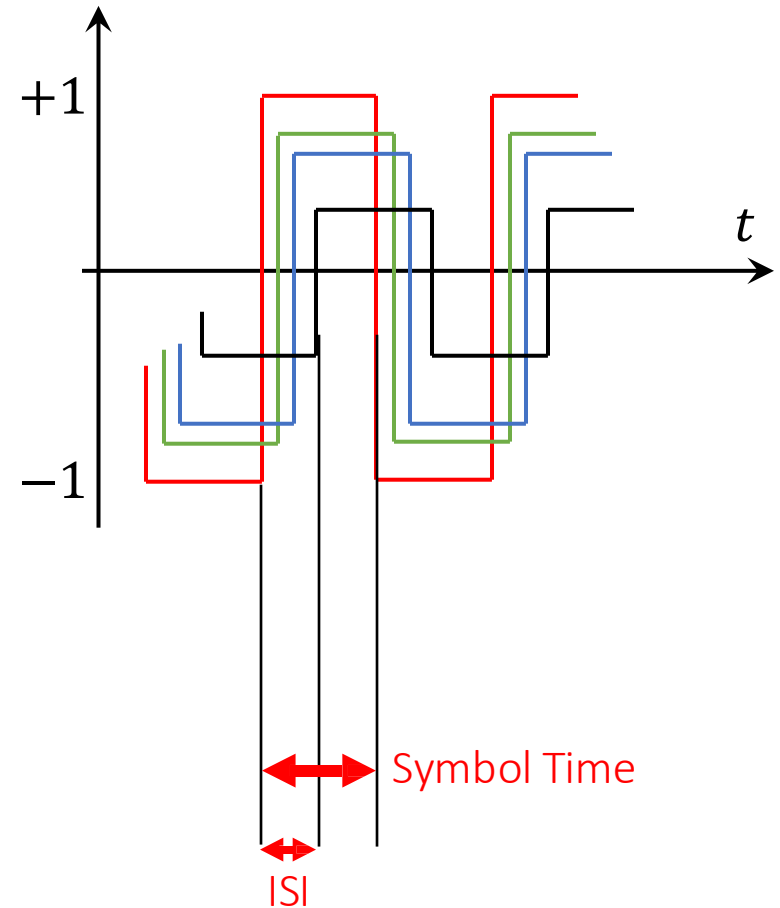
Narrowband Channel

$$h x(t)$$



Wideband Channel

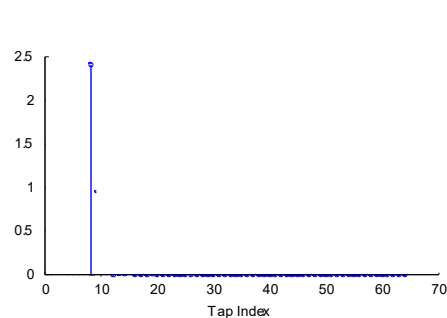
$$h(t) * x(t)$$



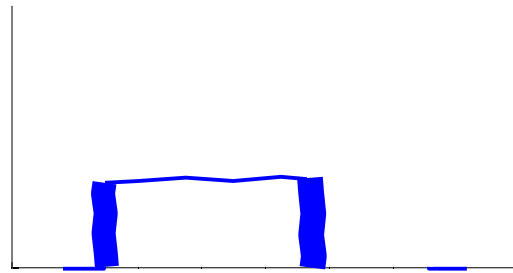
Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t) \iff h X(f)$$



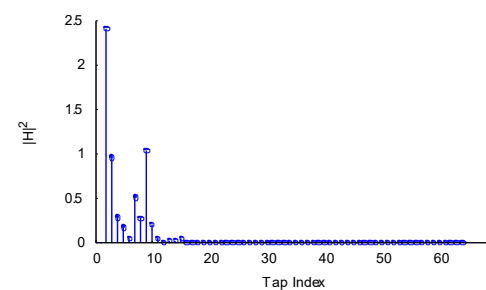
$\approx$ Single Tap



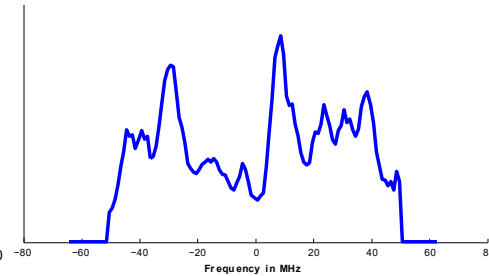
$\approx$ Flat Channel

Wideband Channel

$$h(t) * x(t) \iff H(f) X(f)$$



Multi-tap
Channel
(ISI)



Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: OFDM